VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Project Phase-II report
on
"Pneumonia Detection and Classification using Machine Learning"

Submitted in partial fulfilment of requirement for the "Project Work [BCS786]" of 7th Semester Bachelor of Engineering in Computer Science and Engineering 2025-2026

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DECLARATION

We **SUNIL[1CC22CS098], THANUSHREE U S [1CC22CS104], THRUPTHI H M DODDAMANE[1CC22CS105], VEERESH[1CC22CS109]** declare that this project report titled **"Pneumonia Detection and Classification using Machine Learning"** submitted in partial fulfillment of the degree of B.E in Computer science and engineering, is a record of original work carried out by us under the supervision of **Rekha R**, and has not formed the basis for the award of any other degree, in this or any other institution or university. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been citied.

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ABSTRACT

The infectious illness known as Pneumonia is regularly a result of contamination because of a bacterium in the alveoli of the lungs. While an infected tissue of the lungs has an infection, it builds up pus in it. To find out if the patient has those illnesses, professionals perform bodily exams and diagnose their patients through Chest X-ray, ultrasound, or biopsy of lungs. Misdiagnosis, erroneous treatment, and if the disease is overlooked will result in the patient's lack of lifestyle. The progression of Deep learning contributes to aid in the decision-making procedure of specialists to diagnose sufferers with these illnesses. The look employs a bendy and efficient technique of deep learning of applying the model of CNN in predicting and detecting a patient unaffected and affected with the sickness using a chest X-ray photograph. The take a look at utilizing an accrued dataset of 20,000 images using a 224x224 photograph decision with 32 batch length is applied to prove the overall performance of the CNN model being educated. The trained-version produced an accuracy charge of 95% at some point of the overall performance training. Based on the end result of the experiment carried out, the research study can detect and are expecting COVID-19, bacterial, and viral pneumonia sicknesses based totally on chest X-ray photos.

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CHAPTER 1

INTRODUCTION

This report is the proposal for the "A Machine Learning approach to detection of Pneumonia by using medical images" research project.

The Pneumonia is an illness which inflame the tissues of the one or both lungs. It can be caused by either bacteria or virus like COVID-19. The Pneumonia can be bacterial, viral, aspiration, fungal or hospital-acquired type. Infants, children, old people (above 65 years), smokers, people with heart or kidney or liver diseases and people with weak immune systems are the identified risks groups for the pneumonia disease. Also, it has different severe complications like Septicemia and pleurisy etc.

The WHO and UNICEF gave the "Silent killer of children" and "The forgotten killer" acronyms for the Pneumonia as it is one was the cause for the child (below age 5) mortality, 15% in 2017. This disease kills more children than any other communicable disease, taking lives of more than 800,000 children below age 5 at every year, approximately 2,200 daily. These counts included153, 000 of new-born lives, which implies that a child dies of pneumonia every 39 seconds. The WHO data shows that Sri-Lanka had 6% of child deaths due to the Pneumonia in 2017. The statistics shows the highest child pneumonia mortality and infection rates are distributed around the South Asia and West and Central Africa regions. UNICEF along with WHO has introduced a framework called "Protect, prevent and treat for Pneumonia" to address this critical issue. (Top 20 Pneumonia Facts—2019, 2021), (Pneumonia, 2021), (Ending preventable child deaths from pneumonia and diarrhoea by 2025 - The integrated Global Action Plan for Pneumonia and Diarrhoea (GAPPD), 2013)

Apart from the physical examination of the patients' lungs based on the sounds etc., there are other examine technologies. The Radiography, CT-scan, lung ultrasounds or MRI is the common methods of medical imaging technologies that can be utilized to discover pneumonia clearly among the Pneumonia suspects. Medical personnel check the patient's medical images of their chest to determine if they are infected with pneumonia or not and then take the necessary actions. (Yan, Hui, Lijuan and Zhou, 2019)

From the past Pneumonia still remains a life-threatening disease if not detected or diagnosed earlier. People can find different solutions either to protect, prevent or treat. Non-medical personnel, like Computer Science researchers can develop different Computer Aided Systems (CAD) as supportive tools to diagnose and treatment process.

In the world, the human based analysis of the medical images of patients used to detect the Pneumonia disease. Medical personnel check the patterns or spots of the medical images and then diagnose. Availability or the level of the disease can be imprecise and misinterpreted with another illness due to the subjective errors like experience level, fatigue and biasness etc. A misinterpretation can be causes for the serious complications or even for the death. Also, if the patients are in large numbers the detection will be a time-consuming effort. If this analyzing, detection and diagnosing process can be automated on computerized system, the accuracy of the final decision will be unbiased and without subjective error. And will not be time consumed effort with respect to human based approach. The outcome can achieve a greater level of effectiveness and efficiency.

Computer Aided System (CAD) is an applicable tool in use nowadays for diagnosis and classification of diseases in medical imaging. In achieving the accurate diagnosis, the medical personnel integrate the CAD to assist and verify to support their decision making. Significant features of the images are valuable in using Machine Learning (ML) techniques in this system compared to the traditional hand-crafted features which have boundaries in extracting significant features. The technological advancement in the present day reached a new step closer in human intelligence. It endorses the solutions to solve real-life problems. The Artificial Intelligence (AI) by means of Convolutional Neural Networks (CNN) has capacity in finding substantial features in image classification tasks and provides medical promising results in image analysis. CNN advantages are capable of assisting the identification of some features from an image and use this feature to generate probabilities in classifying specific input. The contribution of this research project is developed as an optimized Machine Learning model of CNN that can detect and classify pneumonia diseases efficiently.

Pneumonia impacts all the elderly and young people everywhere. With the high growth in the popularity of neural networks, engineers and researchers have been able to find state-of-the-art products for computer vision. Artificial Intelligence helps us to automate analysis techniques, which is only possible now because of the technology of Deep Learning. Exposure to Pneumonia is quite high for many people, mainly in economically underdeveloped and developing countries where the majority are deprived of a nutritious diet. The World Health Organization states that more than 4 million untimely deaths per year occur from diseases caused by air pollution.

The purpose of this project is to build an AI network, which takes the pixel values as input for a given X-Ray image and then proceeds to perform linear operations and activations on each of them. Then by taking all the above operations, and then multiplying them with each layer within

the neural network, and the number of nodes. Suddenly you have millions of operations. By applying determination, they can be more efficient in these tasks.

The scope is to develop a model that will identify whether a patient is having Pneumonia or not bypassing the chest X-Ray images through the Deep Learning model.

The global outbreak of the coronavirus disease (COVID-19) in December 2019 has profoundly impacted public health, necessitating the development of rapid and accurate diagnostic methods. Early detection of COVID-19 is crucial for isolating positive cases and preventing the spread of the virus. Traditional diagnostic approaches, such as reverse transcription polymerase chain reaction (RT-PCR) suffer from delayed detection times and varying accuracy, while imaging techniques like X-rays and CT scans offer a faster and more detailed understanding of pulmonary conditions.

Artificial intelligence (AI), especially machine learning (ML) and deep learning (DL), has played an increasingly pivotal role in advancing medical diagnostics as well as in other scientific domains, notably in materials science. Among these techniques, convolutional neural networks (CNNs) have demonstrated significant potential in processing medical images for disease detection. Several studies have shown that CNN-based models can effectively diagnose COVID-19 from radiographic images, such as X-rays and CT scans. However, despite these advances, the secure handling of medical data remains a critical concern, a challenge that can be addressed through federated learning, which allows model training on decentralized data without compromising patient privacy Despite the promise of CNN-based diagnostic tools, existing methods continue to face notable challenges, particularly regarding the ability to distinguish COVID-19 from other pulmonary conditions such as viral pneumonia. These challenges are often exacerbated by limited dataset diversity, slow inference times, and difficulties in distinguishing between similar conditions. Traditional deep learning models rely heavily on large volumes of labeled data and substantial computational resources, which can limit their applicability in certain healthcare environments. To address these issues, hybrid models combining deep feature extraction through CNNs and classification with traditional machine learning algorithms, such as support vector machines (SVMs) have been proposed. Although these models show promise, they still face challenges related to generalization, hyperparameter tuning, and optimization constraints.

In this study, we propose a novel approach that integrates deep learning and transfer learning to enhance the accuracy and efficiency of automatic pulmonary condition detection from radiographic images.

Specifically, we leverage ResNet18, a well-established deep CNN architecture known for its high performance in image classification tasks, and combine it with SVM for the classification of lung conditions. By utilizing transfer learning, we aim to improve model performance, reduce training time, and address challenges associated with limited data availability.

Our approach enables more accurate differentiation between COVID-19, viral pneumonia, lungopacity, and healthy lungs, ultimately contributing to improved diagnostic outcomes. This study introduces a diagnostic framework that merges ResNet18-based deep feature extraction with SVM classification, demonstrating considerable accuracy improvements when employing transfer learning. Furthermore, we address the challenge of data privacy by suggesting federated learning as a viable solution to mitigate privacy concerns in medical AI. The remainder of this paper outlines our methodology, presents the experimental results, and discusses the broader implications of our findings. We conclude with suggestions for future work, including expanding dataset diversity through augmentation techniques and establishing collaborations with clinical experts to further improve the accuracy and applicability of automatic radiographic image detection in medical settings.

Our research aims to advance the detection and differentiation of COVID-19, viral pneumonia, and lung opacity from healthy lungs.This study underscores the potential of combining DL and ML techniques for designing accurate and computationally efficient diagnostic tools, contributing to improved healthcare outcomes for pulmonary diseases. Notably, our approach aligns with the work presented in "Diagnosis of COVID-19 CT Scans Using Convolutional Neural Networks" (SN Computer Science, 5(5), 625), which demonstrates the efficacy of CNN-based models like ResNet152 and DenseNet201 for diagnosing COVID-19 through CT scans. However, in contrast to that study, our method applies ResNet18 and SVM for enhanced accuracy, specifically tailored for radiographic images, which are more widely accessible in clinical settings. This study also explores hyperparameter tuning and implement techniques to address data imbalance, aspects that are critical for improving model reliability. The following sections will outline our methodology, present our findings, discuss their implications, and propose future directions for enhancing automatic radiographic image detection in the medical field.

Section (I) describes the dataset, pretreatment methods, models and statistical analysis conducted in this study. Additionally, we provide a detailed background of architectural adjustments on the CNN, reset 18, VGG, SVM and transfer learning between Reset 18- SVM.

The objectives of this project are to develop a machine learning model that can accurately detect pneumonia from chest X-ray images, while addressing the challenges of class imbalance, variability in image quality, limited generalizability, interpretability, regulatory and ethical considerations, clinical validation, model maintenance, integration with clinical workflow, and liability and accountability.

The model should be able to handle class imbalance in the dataset, where the number of pneumonia cases is significantly lower than non-pneumonia cases. It should also be able to handle variability in image quality, including differences in patient positioning, exposure settings, and equipment used.

The model should be able to generalize well to other datasets and real-world scenarios, including different patient demographics and image acquisition protocols. It should provide interpretable and explainable results, allowing clinicians to understand the reasoning behind the model's predictions.

The model should meet regulatory and ethical standards, including ensuring the model's safety, efficacy, and transparency, as well as addressing potential biases and ensuring patient consent. It should be clinically validated to ensure its accuracy and reliability in real-world settings.

The model should be able to be regularly updated and maintained to ensure continued accuracy and reliability. It should be able to be integrated into clinical workflows, including system compatibility, user interface design, and clinician adoption.

The model should address liability and accountability concerns, particularly in cases where the model's predictions are incorrect or lead to adverse outcomes. The model's performance should be evaluated using key performance indicators (KPIs) such as accuracy, sensitivity, specificity, area under the curve (AUC), and clinical adoption.

The deliverables of this project include a machine learning model that can accurately detect pneumonia from chest X-ray images, a model evaluation report, a clinical validation report, an integration plan, and training and support for clinicians and healthcare professionals on the use of the model.

CHAPTER 2

LITERATURE SURVEY

This chapter will demonstrate the related work and a brief summary of them. The literature review was carried out under three sections namely, accuracy of human based analysis for Pneumonia detection, different medical imaging techniques used in Pneumonia diagnosis and Machine Learning approaches to detect Pneumonia from X-Ray images. Also, future trends discussed.

Venecia Fernandes et al. Proposed, developing a reliable version to detect COVID-19 from X-ray images offers a number of challenges. First of all, there are a limited number of images to be had. The pandemic has simplest been spreading for a few months and now not many datasets have been gathered and shared for the benefit of researchers. Secondly, there may be an urgent need for scientific researchers to develop a wise device for the fast detection and deep expertise of infectious diseases from chest X-ray pictures, which could illustrate the damage to the human body. After investigating the presently available technology and the demanding situations we are going through, it's concluded that a transfer learning of technique is possible and realistic for this study's problem. Our research technique is to train deep learning models at the available images of pneumonia, and contamination in a single or both lungs which may be resulting from microorganisms, viruses, or fungi. The infection causes inflammation inside the air sacs in the lungs, which might be known as alveoli. The alveoli fill with fluid or pus, making it hard to respire the system takes benefit of transfer learning on properly-studied of deep learning to know the type of virus and validates the models on a big quantity of data sets, and finally transfers the models and insights learned in training and validation to a brand new data set in a comparable area, which, in this have a look at, is a new infectious fast-growing sickness, COVID-19 or coronavirus. The proposed technique solves the tough problem in deep learning, this is, how to construct a dependable model primarily based on a scarce information set, which is not well studied and there are unknown features inside the statistics set.

Hongren Lu et al.Deep CNNs indeed acquire higher performance with the use of a huge dataset as compared to a smaller one. Granting there is an enormous range of infected COVID-19 patients globally, however, the variety of publicly available chest X-ray photos online is insignificant and dispersed. For this reason, the authors of this work have defined a reasonably huge dataset of COVID-19 infected chest X-ray images even though normal and pneumonia photographs are right away on hand publicly and applied in this study.

Sammy V. Militante et al. **[3]** Proposed,The observations employ flexible and efficient approaches of deep learning by making use of six models of CNN in predicting and spotting whether the patient is unaffected or affected by the disease employing a chest X-ray picture.

Google Net, Le Net, VGG-16, Alex Net, Strident, and ResNet-50 models with a dataset of 28,000 images and using a 224x224 decision with 32 and sixty-four batch sizes are implemented to verify the performance of every version being educated. The study likewise implements Adam as an optimizer that continues an adjusted 1e-four learning rate and an epoch of 500 hired to all the models. Both Google Net and Le Net acquired a ninety-eight% price, VGG Net -sixteen earned an accuracy charge of ninety-seven%, Alex Net and Stride Net model acquired a ninety-six% whilst the ResNet-50 version acquired 80% throughout the training of models. Google Net and Le Net fashions performed the best accuracy rate for overall performance training. The six models identified had been capable of hitting upon and are expecting pneumonia sickness including a wholesome chest X-ray.

Nanette V. Dionisio et al.Proposed, classify the three varieties of X-rays, image writer used the ensemble method at some stage in prediction, each picture is surpassed through the type layer where they checked whether an image is COVID-19, pneumonia, or ordinary.

Sr. no	Paper Allocation	Author	Year	Methods
1	Bayesian convolutional neural network estimation for paediatric pneumonia detection and diagnosis	Vande cia Fernan des et al.	2021	Bayesian convolutional neural network
2	Transfer Learning from Pneumonia to COVID-19	Hongen Lu et al.	2020	Transfer Learning
3	Pneumonia and COVID-19 Detection using Convolutional Neural Networks	Sammy V. Militante et al.	2021	Convolutional Neural Networks
4	Pneumonia Detection through Adaptive Deep Learning Models of Convolutional Neural Networks	Nanette V. Dionisio et al.	2021	Deep Learning Models

Table 2.1

2.1 Existing system

2.1.1 Accuracy of human based analysis for Pneumonia detection

The researchers had done several studies to analyse the accuracy of doctors in Pneumonia detection and diagnose by using medical images like X-Rays.

The research done by Rao A., shows that there is a gap of accuracy when doctors perform the diagnosis. 24% of doctors find it difficulty in diagnosis X-Ray images. The doctors had mentioned that in order to better diagnosis, they need additional information like Oxygen saturation level, age, respiratory rate etc.

Another study shows the accuracy level of radiologists in Pneumonia detection chest X-Rays with a comparison of a developed machine learning model. Dense Net CNN model with 121 layers had used as the machine learning approach. The radiologists obtained 88.4% of accuracy while the Machine Learning model obtained 96.2% accuracy rate.

2.1.2 Different medical imaging techniques used in Pneumonia diagnosis

The researchers investigate the level of accuracy of the medical images that used in Pneumonia diagnosis. The studies carried by Yan C. 2019 done the comparison with Lung ultrasound, CT scans & X-Ray medical imaging techniques. Its shows that highest accuracy was obtained by CT scans, then ultra-sound and lastly X-Ray images. Also, they had mentioned that CT scans and ultrasound is expensive and not commonly used with respect to the X-Rays.

Another research team had performed research on chest ultrasound Vs X-Ray accuracy in Pneumonia diagnosis. The ultrasound had 66.1% accuracy rate and X-Ray had 55.9% accuracy rate. (The significance of chest ultrasound and chest X-ray in the diagnosis of children clinically suspected of pneumonia, 2021).

2.1.3 Machine Learning approaches to detect Pneumonia from X-Ray images

There are many researches done to evaluate the Machine Learning approaches in detection and diagnosis of Pneumonia. They have evaluated different Machine Learning algorithms like decision tress, logistic regression etc. Also, the evaluation of the pre-trained CNN models available. The research done by Rao A. had evaluate the accuracy of four machine learning algorithms Nearest Neighbour, Decision tree, Logistic regression and CNN. The accuracy levels are 70%, 66%, 67%.

Another study shows different accuracy levels obtained by the pre-trained CNN models. They have used the transfer learning and fine tuning on the Exception model with a slight change, adding two fully connected layers and two-way output layer with SoftMax function after the global average polling layer. For VGG16, they have done a change to the last 8 early layers to avoid the train. Exception model obtained 82%, while Vgg16 model obtained 87% of accuracy. Also, it evaluates the performances of the used two CNN models. Xception model obtained trained time as 100 minutes and test time as 13 seconds, while Vgg16 model obtained of obtained trained time as 83 minutes and test time 10 seconds.

The research done by using the ResNet-50 and DenseNet-161 models to implement a customized deep network model to increase the pneumonia detection accuracy. They also evaluated the currently available systems. They have used transfer learning, data augmentation. In DenseNet they have removed the fully connected layer from the transfer learning task. The rest of remaining layers of the model used as the function extractor. The proposed model was trained by ImageNet dataset lo learn visual patterns using the first layer set. Then the complex features learned using the deeper layers. At last, final layers recognize a proper structure by the combination of patterns which was previous learned. They achieved 93% of accuracy. (Alsu Mairi and Ismail, 2020) Another study had developed a hybrid system which is utilizing the Random Forest (RF) and Adaptive Median Filer Convolutional Neural Network (ACNN). They have used different preprocessing methods to remove the noises, Dropout function-based method to extract the deep features. In the last stage the Random Forest classifier had been used as the classifier. This model was obtained 93.8% of accuracy.

Another research creates a model with the aid of none rigid registration driven lung segmentation method. They have focused on lung segmentation as it can make less noises with respect to the full X-Ray image. Their approach consists with feature extraction of eight statistical features that constructed on the first order histogram of the original image, classification steps.

Marwa Eid Yasser H. Elway had developed a Machine Learning model based on the Rest Net based Support Vector Machine (SVM) classification. Their model consisted with pre-processing techniques like resizing and normalization, Rest Net 50 CNN model, Adaptive Boosting (AdaBoost) Algorithm for feature selection and SVM for the classification. This model obtained 98.13 % accuracy.

Another research had proposed a multichannel CNN for detection of Pneumonia from X-Ray images which achieved 97% of accuracy. They have a series of pre-processing functions such as Contrast-Limited Adaptive Histogram Equalization (CLAHE), image sharping, resizing, cropping etc. The solution extracted two different sets of features to offer the learning algorithm two diverse views on the training data. The first feature was CLAHE image-based features and the second was sharpened-Canny image. These two set of feature extractions works as the multichannel of the proposed solution.

Another study had done a great comparison with 6 pre-trained CNN models which utilize the adaptive deep learning. LeNet, GoogLeNet, VGGNet-16, AlexNet, StridedNet, and ResNet-50 models were the models that had been evaluated. Like the previous study, this work utilized the cropping, resizing and data augmentation as the pre-processing stages. The evaluation results shows that 98% of accuracy rate was obtained by the GoogLeNet and LeNet. The ResNet-50 obtained 80%, VGG-16, AlexNet, and StridedNet gained 96% - 97% accuracy rates.

Another research proposes a Deep Convolutional Neural Network (DCNN) with 52 Convolutional layers along with 2 dense layers. They have used data augmentation and different pre-processing techniques for the dataset. The proposed model contains feature-extraction and classification stages. Feature extractor consists with 52 Convolutional layers with Max-Pooling. The achieved accuracy rate for this proposed model was 96.09%. (Yi et al., 2021)

Another research had been developed a CNN based model with transfer learning to perform the similar task with the combination of AlexNet, DenseNet121, InceptionV3, resNet18 and GoogleNet, feature extraction and ensemble classification. Pre-processing, data augmentation also used. To combine the 5 CNN models, they have used ensemble classification. To train the proposed model with the dataset the Adam optimizer and cross-entropy loss function used. 98.9% accuracy rate was achieved in this model.

Another research had evaluated the accuracy of VGG16 CNN model with changing the final classifier. They have used Support Vector Machine (SVM), Random Forest (RF) and logistic regression as the different classifiers. The logistic regression classifier had obtained the highest accuracy of 96.36% rate while SVM and RF classifiers obtained 95.24%, 95.2% respectively.

A combination different CNN models (Alex Net, VGG-16 and VGG-19) and MrMR feature selection with different classifiers (Decision Trees, K-Nearest Neighbour, Support Vector Machine and Linear regression) was evaluated in a study.

After each convolutional layer a batch normalization layer used for the regularization. SoftMax activation function used to classification at the final layer. The model was trained with Adam optimizer.

Research proposed a transfer learning method with residual network. It had obtained 90.5% of accuracy rate. With the feature extractions by aid of DesneNet-169 with SVM classification, another research obtained 80.02% of accuracy rate.

Another study developed a machine learning model with 96.7% accuracy rate. It utilized AlexNet based full connected layer (AN-FCN) CNN model to segmentation of lung regions. Then the CNN model for the classification and extraction of features. The features that were extracted by this CNN model was high dimensional fully connected layers features, Gray Level Co-occurrence Matrix (GLMC), Histogram of Oriented Gradients (HOG) shape features and Haar Wavelet texture features. The extracted feature classification is done by the SVM classifier.

96.76% accuracy rate was achieved a research, ResNet-50 model was used as the CNN model. The CNN model had 49 layers for the feature extraction and introduced one new prediction layer. The new prediction layer contained two nodes.

A performance assessment of five CNN models, AlexNet, GoogleNet, LeNet, ResNet, and Vignette were done by the study and they have found that GoogleNet and LeNet obtained 97% of accuracy. The VGGNet and AlexNet obtained 96% and RestNet obtained 74% off accuracy rate.

2.2 Drawbacks

1. Data Quality and Availability: Machine learning models require large amounts of high-quality data to learn and make accurate predictions. However, collecting and annotating medical images, such as chest X-rays, can be challenging due to factors like data privacy, varying image quality, and the need for expert annotation.

2. Class Imbalance: Pneumonia datasets often suffer from class imbalance, where the number of pneumonia cases is significantly lower than non-pneumonia cases. This can lead to biased models that are more accurate in predicting non-pneumonia cases but less accurate in detecting pneumonia.

3. Variability in Image Quality: Chest X-ray images can vary in quality due to factors like patient positioning, exposure settings, and equipment used. This variability can affect the model's ability to accurately detect pneumonia.

4. Limited Generalizability: Machine learning models trained on one dataset may not generalize well to other datasets or real-world scenarios. This can be due to differences in data distribution, image acquisition protocols, or patient demographics.

5. Interpretability and Explainability: Deep learning models, in particular, can be difficult to interpret and explain, making it challenging to understand why a particular prediction was made. This can be a concern in medical diagnosis, where understanding the reasoning behind a prediction is crucial.

6. Regulatory and Ethical Considerations: The use of machine learning in medical diagnosis raises regulatory and ethical concerns, such as ensuring the model's safety, efficacy, and transparency, as well as addressing potential biases and ensuring patient consent.

7. Clinical Validation: Machine learning models require clinical validation to ensure their accuracy and reliability in real-world settings. This can be time-consuming and resource-intensive, requiring collaboration between clinicians, data scientists, and regulatory bodies.

8. Model Drift and Maintenance: Machine learning models can drift over time due to changes in data distribution, patient demographics, or image acquisition protocols. Regular model updates and maintenance are necessary to ensure continued accuracy and reliability.

9. Integration with Clinical Workflow: Machine learning models need to be integrated into clinical workflows, which can be challenging due to factors like system compatibility, user interface design, and clinician adoption.

10. Liability and Accountability: The use of machine learning in medical diagnosis raises questions about liability and accountability, particularly in cases where the model's predictions are incorrect or lead to adverse outcomes.

2.3 Proposed System

In this study, we designed a CNN model to extract the features of chest X-ray images and use those features to detect if a patient suffers from pneumonia. In Our CNN Architecture, we began it with a lower filter value of 32 and increased it layer-wise. Constructed the model with a layer of Conv2D followed by a layer of Carpooling. The kernel size is preferred to be an odd number like 3x3.

Tanh, ReLU, etc. can be used for activation function, but ReLU is the most preferred activation function. input shape takes in image width & height with the last dimension as a color channel. then we Flatten the input after CNN layers and added ANN layers.

$$S(x) = \frac{1}{1 + e^{-x}}$$

$$f(x)=\max(0, x)$$

$$S(x) = Sigmoid$$

$$f(x) = ReLU$$

Used activation function as SoftMax for the last layers (ANN Layers) also defined units as the total number of classes and used sigmoid for binary classification and set unit to 1.

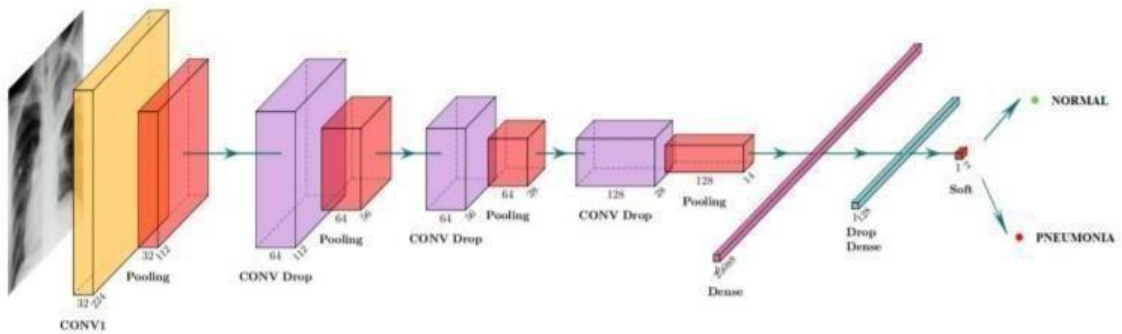


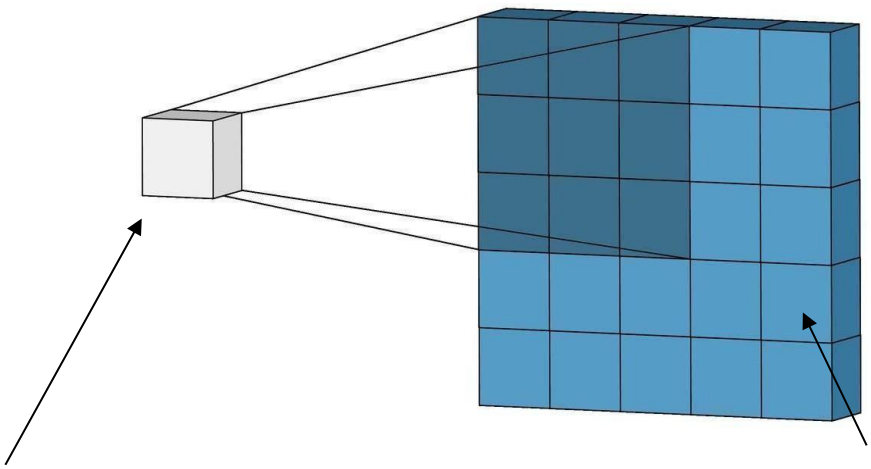
Fig 2.3.1 Details of proposed DL mode

2.4 Working

Convolutional neural networks refer to a sub-category of neural networks: they, therefore, have all the characteristics of neural networks. However, CNN is specifically designed to process input images. Their architecture is then more specific: it is composed of two main blocks.

2.4.1 Conv Layers

The first block makes the particularity of this type of neural network since it functions as a feature extractor. To do this, it performs template matching by applying convolution filtering operations. The first layer filters the image with several convolution kernels and returns “feature maps”, which are then normalized (with an activation function) and/or resized.



Image

Convolved feature
(feature Map)

Fig 2.4.1 Conv Layer

2.4.2 Pool Layers

The second block is not characteristic of a CNN: it is in fact at the end of all the neural networks used for classification. The input vector values are transformed (with several linear combinations and activation functions) to return a new vector to the output. This last vector contains as many elements as there are classes: element I represents the probability that the image belongs to class I. Each element is therefore between 0 and 1, and the sum of all is worth 1. These probabilities are calculated by the last layer of this block (and therefore of the network), which uses a Sigmoid function (binary classification) or a RELU function (multi-class classification) as an activation function.

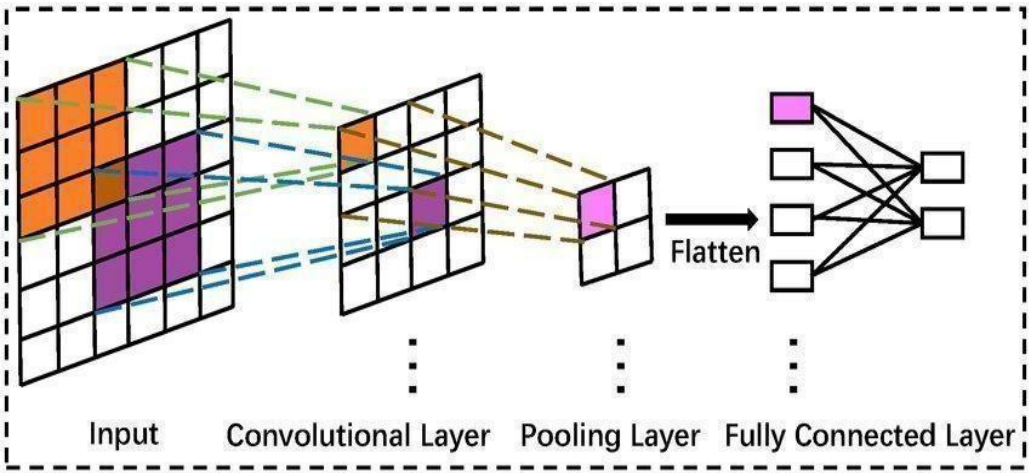


Fig 2.4.2 Mathematics of Conv and Pooling Layer

2.4.3 Activation Function

An activation function is a mathematical function applied to the output of a neuron in a neural network. It introduces non-linearity to the network, allowing it to learn and model complex relationships between inputs and outputs. Common activation functions include the sigmoid, ReLU, and tanh functions.

ReLU

Rectified linear unit is most widely used and preferred activation function right now which ranges from 0 to infinity, All the negative values are converted into zero.

$$f(x)=\max(0, x)$$

2.4.4 Sigmoid

The sigmoid function also called a logistic function. having a characteristic that can take any real value and map it to between 0 to 1. It decides which value to pass as output and what not to pass.

$$S(x) = \frac{1}{1 + e^{-x}}$$

2.4.5 VGG16 Model

VGG16 is a popular convolutional neural network architecture for image classification tasks. It consists of 16 layers, including 13 convolutional layers and 3 fully connected layers. The convolutional layers use small 3x3 filters with a stride of 1 and are followed by max pooling layers. The fully connected layers at the end of the network perform the classification task. The VGG16 model has achieved state-of-the-art performance on several image recognition benchmarks, making it a popular choice for computer vision tasks.

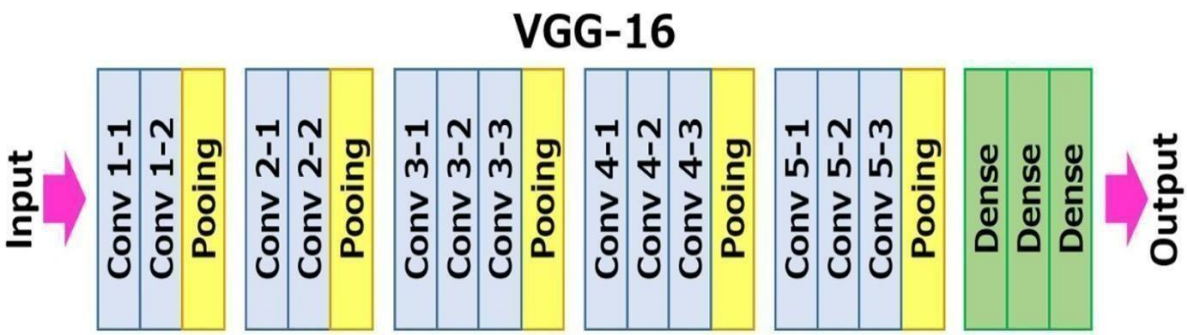


Fig 2.4.3 VGG16 Model Sequence

This chapter will discuss the work schedule of the project. The work plan has task identification, time allocation and budget estimation sections. The time allocation is illustrated using a Gantt chart.

2.4.6 Task identification

Following tasks has been identified as to see the progress of the project. They are listed as

- Ideation and literature survey
- Data gathering
- Data pre-processing on the selected datasets (Grouping, validating and labelling)
- Dataset splitting, data augmentation, pre-processing
- Planning and design of a suitable CNN model
- Implementing the CNN model
- Train the CNN model
- Testing the accuracy of the model

2.4.7 Time allocation

Following Gantt chart will illustrate the time allocation of the project.

Task Name	Start (Date)	End (Date)	Duration (Days)
Ideation and literature survey	7/1/2021	7/19/2021	18
Data gathering	7/20/2021	7/25/2021	5
Dataset pre-processing	7/26/2021	8/10/2021	15
Dataset splitting	8/10/2021	8/12/2021	2
Data augmentation	8/11/2021	8/20/2021	9
Pre-processing	8/20/2021	9/1/2021	12
Planning and design of model	9/1/2021	9/15/2021	14
Implementing the CNN model	9/15/2021	10/20/2021	35
Train the CNN model	10/20/2021	12/1/2021	42
Testing the accuracy of the model	12/1/2021	12/10/2021	9

Table 2.4.1 Time allocation

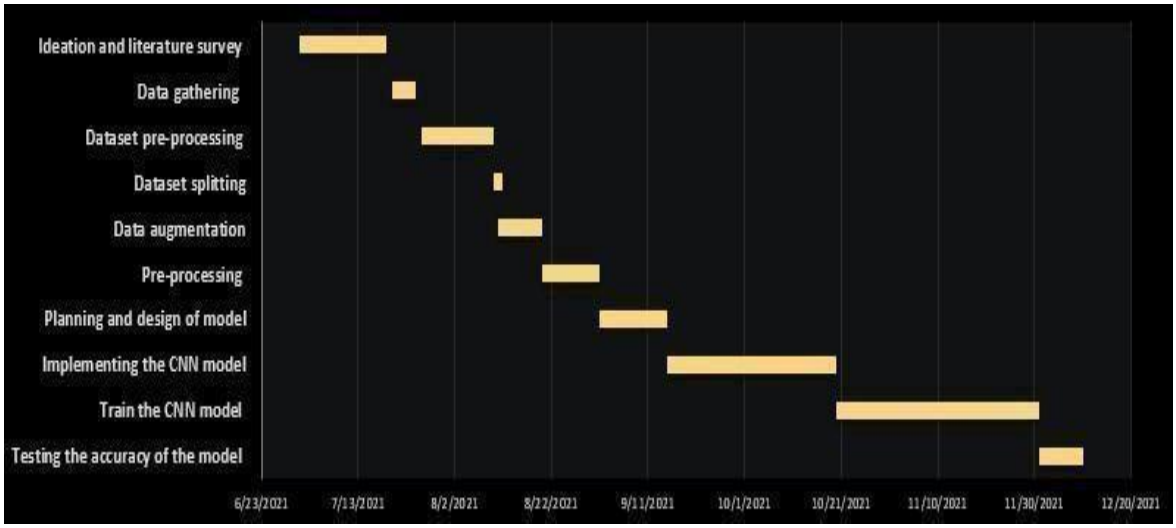


Figure 2.4.4 Gantt chart

Budget allocation

Currently a budget estimation and allocation are not necessary. If there is a requirement to use Google Colab paid services, a budget allocation will be there. Anyhow the project will be selffunded.

Summary

This chapter explained the work plan under the subsections of task identification, time allocation and budget estimation. The next chapter will be discussing the overall overview of this report.

CHAPTER 3

PROBLEM STATEMENT

The human based analysis of the medical images of patients to detect the Pneumonia may contain different subjective errors due to the expertise levels, fatigue and biasness etc. Also, if the patients are in large numbers the detection will be a time-consuming effort. If this analyzing, detection and diagnosing process can be automated on computerized system, the accuracy of the final decision will be unbiased and without subjective error. And will not be time consumed effort with respect to human based approach. This project will propose to develop an accurate and efficient Machine Learning mechanism to automate the detection of Pneumonia by using medical images without subjective errors.

3.1 Consequences

1. Delayed Diagnosis: Without a machine learning model, pneumonia diagnosis may be delayed, leading to delayed treatment and potentially worse patient outcomes.

2. Inaccurate Diagnosis: Human clinicians may misdiagnose pneumonia, leading to incorrect treatment and potentially worse patient outcomes.

3. Increased Healthcare Costs: Delayed or inaccurate diagnosis may lead to increased healthcare costs due to prolonged hospital stays, additional testing, and unnecessary treatments.

4. Reduced Patient Care: Without a machine learning model, clinicians may not be able to provide the best possible care for patients with pneumonia, leading to reduced patient outcomes and satisfaction.

5. Increased Mortality Rates: Delayed or inaccurate diagnosis may lead to increased mortality rates due to pneumonia.

3.2 Proposed Solution

Computer Science filed can actively support to develop different Computer Aided Systems (CAD) as supportive tools to diagnose and treatment process to address this life-threating issue. Hence, the proposed solution is to develop a CAD system that is enable to detect of Pneumonia by using medical images.

The direct users and one of the beneficiaries of this proposed project will be the medical workers, which will be act supportive tool the decision-making process of the Pneumonia detection and diagnosis.

Medical images will act as the input of this solution and the processing will be done via a trained CNN model for diagnose of the Pneumonia. The CNN model along with other classification algorithms and image processing techniques, analyze the image and extract the features. Based on the features the model will generate "1" or "0". '1' will represent the positive Pneumonia case and negative will represent the negative Pneumonia case.

The system will utilize the Machine Learning approach and image processing techniques. The Convolutional Neural Networks will be the Machine Learning approach to be used to the detection and diagnose of the Pneumonia and different image processing techniques like normalization etc. will be used to image pre-processing task.

The proposed solution is needed to have dataset for the training of CNN model. The dataset should be licensed to use and should be adequate in amount. Also, different software and hardware requirements should be available to implement and execute the proposed solution.

3.3 Goals

1. Improve Pneumonia Detection Accuracy: Develop a machine learning model that can accurately detect pneumonia from chest X-ray images, reducing the risk of delayed or inaccurate diagnosis.

2. Enhance Patient Care: Use the machine learning model to provide clinicians with accurate and timely diagnoses, enabling them to provide the best possible care for patients with pneumonia.

3. Reduce Healthcare Costs: By reducing the need for prolonged hospital stays, additional testing, and unnecessary treatments, the machine learning model can help reduce healthcare costs associated with pneumonia diagnosis and treatment.

4. Improve Research: Use the machine learning model to gain a better understanding of pneumonia and develop more effective treatments, leading to improved patient outcomes and reduced mortality rates.

5. Increase Efficiency: Develop a machine learning model that can efficiently process medical images, reducing the time and effort required for manual interpretation and enabling clinicians to focus on more complex and high-value tasks.

6. Enhance Patient Outcomes: By providing accurate and timely diagnoses, the machine learning model can help improve patient outcomes, reducing the risk of complications and mortality associated with pneumonia.

7. Reduce Mortality Rates: By enabling clinicians to provide more effective care and reducing the risk of delayed or inaccurate diagnosis, the machine learning model can help reduce mortality rates associated with pneumonia.

8. Improve Clinician Confidence: By providing accurate and reliable diagnoses, the machine learning model can help improve clinician confidence in their diagnoses, reducing the risk of misdiagnosis and improving patient outcomes.

9. Enhance Patient Satisfaction: By providing accurate and timely diagnoses, the machine learning model can help improve patient satisfaction, reducing the risk of dissatisfaction and improving overall patient experience.

10. Improve Healthcare Quality: By providing accurate and reliable diagnoses, the machine learning model can help improve the overall quality of healthcare, reducing the risk of medical errors and improving patient outcomes.

3.4 Motivation

The goal of this proposed solution is using Machine Learning and Image Processing techniques to identify and diagnose of Pneumonia of patient's chest X-rays. So that the patient can be directed to the treatments as per the requirement. This will be a simple implementation "treat" of the UNICEF & WHO "Protect, prevent and treat for Pneumonia" framework.

1. Handle Class Imbalance: Develop a model that can handle class imbalance in the dataset, where the number of pneumonia cases is significantly lower than non-pneumonia cases.

2. Improve Data Quality: Develop a model that can handle variability in image quality, including differences in patient positioning, exposure settings, and equipment used.

3. Increase Generalizability: Develop a model that can generalize well to other datasets and real-world scenarios, including different patient demographics and image acquisition protocols.

4. Improve Interpretability: Develop a model that provides interpretable and explainable results, allowing clinicians to understand the reasoning behind the model's predictions.

5. Address Regulatory and Ethical Considerations: Develop a model that meets regulatory and ethical standards, including ensuring the model's safety, efficacy, and transparency, as well as addressing potential biases and ensuring patient consent.

6. Clinical Validation: Develop a model that has been clinically validated to ensure its accuracy and reliability in real-world settings.

7. Model Maintenance: Develop a model that can be regularly updated and maintained to ensure continued accuracy and reliability.

8. Integration with Clinical Workflow: Develop a model that can be integrated into clinical workflows, including system compatibility, user interface design, and clinician adoption.

9. Liability and Accountability: Develop a model that addresses liability and accountability concerns, particularly in cases where the model's predictions are incorrect or lead to adverse outcomes.

The problem statement for pneumonia detection is to develop a machine learning model that can accurately detect pneumonia from chest X-ray images, addressing the challenges of class imbalance, variability in image quality, limited generalizability, interpretability, regulatory and ethical considerations, clinical validation, model maintenance, integration with clinical workflow, and liability and accountability.

The goal is to improve pneumonia detection accuracy, enhance patient care, reduce healthcare costs, and improve research. The model should be able to accurately detect pneumonia from chest X-ray images, provide clinicians with accurate and timely diagnoses, and enable them to provide the best possible care for patients with pneumonia.

The deliverables include a machine learning model, a model evaluation report, a clinical validation report, an integration plan, and training and support for clinicians and healthcare professionals on the use of the model. The model should be able to generalize well to other datasets and real-world scenarios, and be interpretable, transparent, and accountable.

The problem statement also highlights the need for clinical validation, model maintenance, and integration with clinical workflow, as well as the importance of addressing regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

Overall, the problem statement emphasizes the need for a machine learning model that can accurately detect pneumonia from chest X-ray images, while addressing the challenges and considerations associated with developing and deploying such a model in a clinical

The development of a machine learning model for pneumonia detection has the potential to significantly impact the diagnosis and treatment of pneumonia, a disease that affects millions of people worldwide each year. Pneumonia is a leading cause of morbidity and mortality, particularly among vulnerable populations such as the elderly and young children.

The use of machine learning in pneumonia detection has the potential to improve patient outcomes, reduce healthcare costs, and enhance the overall quality of care. However, the development and deployment of such a model also raise important regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

The problem statement highlights the need for a comprehensive approach to developing a machine learning model for pneumonia detection, one that addresses the technical, clinical, and regulatory challenges associated with this task. This includes the need for high-quality training data, robust model evaluation and validation, and careful consideration of the model's potential impact on patient care and outcomes.

The development of a machine learning model for pneumonia detection is a complex task that requires the collaboration of clinicians, data scientists, and regulatory experts. It is a task that requires careful consideration of the technical, clinical, and regulatory challenges associated with developing and deploying such a model, as well as a commitment to ensuring the model's safety, efficacy, and transparency.

In conclusion, the problem statement for pneumonia detection emphasizes the need for a machine learning model that can accurately detect pneumonia from chest X-ray images, while addressing the challenges and considerations associated with developing and deploying such a model in a clinical setting. The development of such a model has the potential to significantly impact the diagnosis and treatment of pneumonia, and requires a comprehensive approach that addresses the technical, clinical, and regulatory challenges associated with this task.

The problem statement also highlights the need for ongoing research and development in the field of machine learning for pneumonia detection, as well as the importance of collaboration between clinicians, data scientists, and regulatory experts. By working together, we can develop a machine learning model that can accurately detect pneumonia from chest X-ray images, and improve patient outcomes, reduce healthcare costs, and enhance the overall quality of care.

Furthermore, the problem statement emphasizes the need for a machine learning model that can generalize well to other datasets and real-world scenarios, and be interpretable, transparent, and accountable. This includes the need for careful consideration of the model's potential impact on patient care and outcomes, as well as the importance of addressing regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

In addition, the problem statement highlights the need for clinical validation, model maintenance, and integration with clinical workflow, as well as the importance of addressing regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

Overall, the problem statement for pneumonia detection emphasizes the need for a comprehensive approach to developing a machine learning model for pneumonia detection, one that addresses the technical, clinical, and regulatory challenges associated with this task. This includes the need for high-quality training data, robust model evaluation and validation, and careful consideration of the model's potential impact on patient care and outcomes.

The development of a machine learning model for pneumonia detection is a complex task that requires the collaboration of clinicians, data scientists, and regulatory experts. It is a task that requires careful consideration of the technical, clinical, and regulatory challenges associated with developing and deploying such a model, as well as a commitment to ensuring the model's safety, efficacy, and transparency

CHAPTER 4

OBJECTIVES

To learn different biomedical terms related to pneumonia disease.

To learn the different scenarios of pneumonia disease (viral or bacterial).

To learn different methods of data acquisition. to know about different image processing pre-trained models.

To build a web application to analyse Pneumonia disease using chest X-ray.

To analyse the potential use of computer vision and machine learning models to address Pneumonia detection

To develop a machine learning model for image classification

To develop a fast and efficient Pneumonia diagnosis system without subjective error in the process of diagnosis

1. Develop a Machine Learning Model: Create a machine learning model that can accurately detect pneumonia from chest X-ray images.

2. Address Class Imbalance: Develop a model that can handle class imbalance in the dataset, where the number of pneumonia cases is significantly lower than non-pneumonia cases.

3. Improve Data Quality: Develop a model that can handle variability in image quality, including differences in patient positioning, exposure settings, and equipment used.

4. Increase Generalizability: Develop a model that can generalize well to other datasets and real-world scenarios, including different patient demographics and image acquisition protocols.

5. Improve Interpretability: Develop a model that provides interpretable and explainable results, allowing clinicians to understand the reasoning behind the model's predictions.

6. Address Regulatory and Ethical Considerations: Develop a model that meets regulatory and ethical standards, including ensuring the model's safety, efficacy, and transparency, as well as addressing potential biases and ensuring patient consent.

7. Clinical Validation: Develop a model that has been clinically validated to ensure its accuracy and reliability in real-world settings.

8. Model Maintenance: Develop a model that can be regularly updated and maintained to ensure continued accuracy and reliability.

9. Integration with Clinical Workflow: Develop a model that can be integrated into clinical workflows, including system compatibility, user interface design, and clinician adoption.

The primary objective of this project is to develop a machine learning model that can accurately detect pneumonia from chest X-ray images. This objective can be achieved by improving pneumonia detection accuracy, enhancing patient care, reducing healthcare costs, improving research, increasing efficiency, enhancing patient outcomes, reducing mortality rates, improving clinician confidence, enhancing patient satisfaction, and improving healthcare quality.

The model should be able to accurately detect pneumonia from chest X-ray images, provide clinicians with accurate and timely diagnoses, and enable them to provide the best possible care for patients with pneumonia. The model should also be able to efficiently process medical images, reducing the time and effort required for manual interpretation and enabling clinicians to focus on more complex and high-value tasks.

The development of a machine learning model for pneumonia detection has the potential to significantly impact the diagnosis and treatment of pneumonia, a disease that affects millions of people worldwide each year. Pneumonia is a leading cause of morbidity and mortality, particularly among vulnerable populations such as the elderly and young children.

The use of machine learning in pneumonia detection has the potential to improve patient outcomes, reduce healthcare costs, and enhance the overall quality of care. However, the development and deployment of such a model also raise important regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

The model should be able to generalize well to other datasets and real-world scenarios, and be interpretable, transparent, and accountable. This includes the need for careful consideration of the model's potential impact on patient care and outcomes, as well as the importance of addressing regulatory and ethical considerations, such as ensuring the model's safety, efficacy, and transparency, and addressing potential biases and ensuring patient consent.

4.1 Key Performance Indicators (KPIs):

The model's performance will be evaluated using several key performance indicators (KPIs), including accuracy, sensitivity, specificity, area under the curve (AUC), and clinical adoption. These KPIs will provide a comprehensive understanding of the model's ability to detect pneumonia from chest X-ray images and its potential impact on patient care and outcomes.

4.1.1 Deliverables:

The deliverables for this project include a machine learning model that can accurately detect pneumonia from chest X-ray images, a model evaluation report, a clinical validation report, an integration plan, and training and support for clinicians and healthcare professionals on the use of the model.

4.1.2 Timeline:

The project timeline includes data collection and preprocessing, model development and training, model evaluation and validation, clinical validation and integration planning, and deployment and maintenance. The project is expected to be completed within 12 months.

4.1.3 Resources:

The project requires a large dataset of chest X-ray images with corresponding labels (pneumonia or non-pneumonia), high-performance computing hardware and software for model training and evaluation, clinical expertise, and regulatory expertise. The project also requires collaboration with clinicians, data scientists, and regulatory experts to ensure the model's accuracy, reliability, and compliance with relevant laws and regulations.

4.1.4 Improved Patient Outcomes: By providing accurate and timely diagnoses, the machine learning model can help improve patient outcomes, reducing the risk of complications and mortality associated with pneumonia.

4.1.5 Reduced Healthcare Costs: By reducing the need for prolonged hospital stays, additional testing, and unnecessary treatments, the machine learning model can help reduce healthcare costs associated with pneumonia diagnosis and treatment.

4.1.6 Enhanced Patient Satisfaction: By providing accurate and timely diagnoses, the machine learning model can help improve patient satisfaction, reducing the risk of dissatisfaction and improving overall patient experience.

CHAPTER 5

METHODOLOGY

Machine learning is primarily used through traditional approaches to detect and classify various anomalies. The main issue with this approach is the excessive computational burden and its limited performance. In this section, we provide a detailed description of the data, analysis techniques, as well as data collection and annotation, as shown in Fig. 1. We then discuss the training and design of CNN architectures and transfer learning.

COVID-19 X-ray database

The purpose of this approach is to detect and classify COVID-19, viral pneumonia, and lung opacity on the chest X-ray images similar to the samples.

5.1 Data augmentation

In deep learning, it is often desirable to implement precise data augmentation to avoid overfitting issues. This can also be mitigated through a well-designed data growth strategy [20,21]. The latter can be developed to improve sample size, thus reducing the impact of imbalanced data. Data augmentation is a strategy that addresses the issue of imbalanced data by introducing slight variations to existing data [22, 23]. For example, Soumya Ranjan Nayak et al. augmented the training of X-ray images by rotating them clockwise by 5°, scaling them by 15 %, flipping the images horizontally, and adding Gaussian noise with a mean of 0 and a variance of 0.25 [24] as shown in Fig. 3 (a).

These methods do not generate new visual features in the images that could enhance the overall learning capabilities of the algorithm used and the network's generalization capabilities. Similarly, color, texture, and geometry-based augmentation techniques are not as popular due to their various drawbacks. Although a wide variety of other interesting methods have been developed in the past, only geometric transformations are currently commonly used. In this study, we used image rotation for geometric transformation to augment the chest X-ray images. While this technique allows for data diversification and improves classification, it also has drawbacks, such as additional memory, conversion, and computational costs, as well as extra learning time [25]. The rotation augmentations have been applied to images belonging to the four classes shown in Table 1, which contain a smaller amount of data.

5.2 CNN architecture

In this work, CNN, which is a cutting-edge sphere of machine learning inspired by the human brain, was used to classify images into COVID-19, normal, viral pneumonia and lung opacity. Fig. 3 (b) summarizes the steps of the CNN architecture for the differentiation of the different classes. Here,

the CNN functions similarly to a human visual system, making it well-suited for two-dimensional images. This approach is particularly effective for medical image analysis, as demonstrated by Dhiman et al. [26], who achieved a classification accuracy of 98.54 % for COVID-19 detection using ResNet101 and the J48 algorithm. Thus, CNN analyzes images with pixel patterns to extract feature maps. However, the units are related to the previous layers from the core weights which are modified during formation through a process of backpropagation. Since all units use similar kernels, less weight is generated per convolutional layer. Indeed, the flats produced with CNN to achieve the objective of classifying the thoracic image are as following equation:

$f(x)=\max(0, x)$

Where, f(x) is the activation function to make the data on a linear unit in range 0, x. The pooling eliminated some important elements, thus leading to overlapped pooling 3*3. In conventional layers, as per eq. (2) functionality, the cards were padded with convolution. The use of padding ensured function cards of the same dimensions. Fig. 4 (a) represents the hyper-parameters as feature selection and optimization in achieving high performance in image classification tasks.

5.3 Machine learning: SVM model

Support Vector Machine (SVM), as shown in Fig. 4 (b), allows the computer to learn how to perform classification and regression tasks, using an algorithm based on statistical learning and optimization theory, which increases the accuracy of predictions and completely avoids the

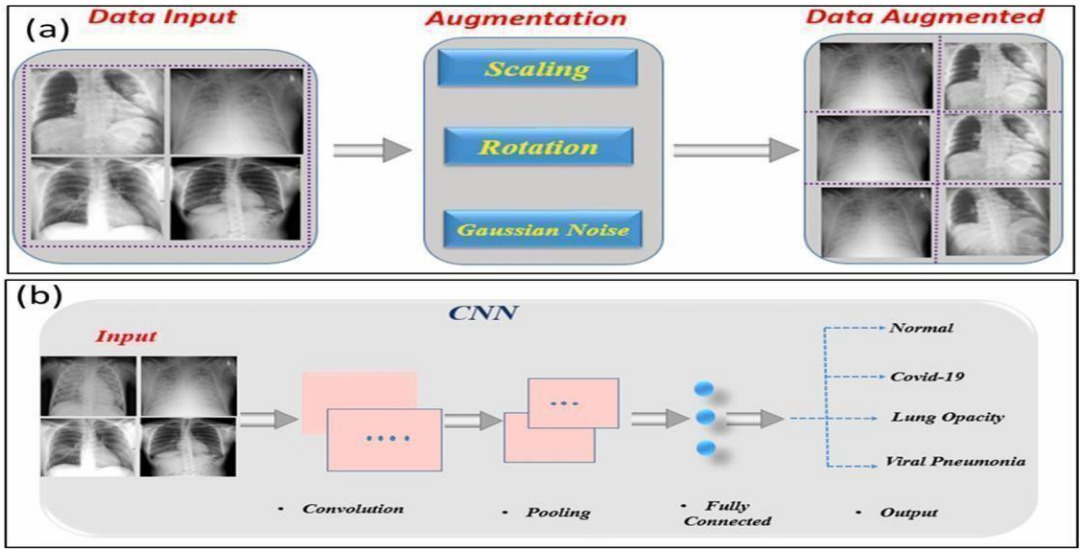


Fig. 5.1 (a) Sample results of data augmentation, (b) A basic CNN architecture for the multi-classification of COVID-19, Normal, Viral pneumonia and Lung Opacity.

The number of samples belonging to each category in the COVID-19 radiographic database.

Category	Number of images
COVID-19	3616
Viral Pneumonia	1345
Normal	10192
Lung Opacity	6012
Total	21165

Table 5.1

disadvantages of over-modification. However, classifiers utilize a geometric representation of the data, where data points are plotted based on their characteristics or variables that influence classifications The role of the SVM is to define a decision boundary (linear decision area) that separates datasets with the largest deviation (maximum margin) on each side according to the optimization technique that can be solved using quadratic programming. In this study, we used one of the most widely known kernel functions, the Gaussian kernel (also known as the Radial Base Function (RBF)) to enhance classification performance.

5.4 Data Acquisition

The proposed database used to evaluate the performance of the model contains a total of 5863 X-ray photos from the Kaggle.

In 2017 Dr. Paul Mooney started a competition on Kaggle on viral and bacterial pneumonia classification. It contained 5,863 pediatric images; hence it is very different from the other datasets. We are referring to the revised version of this dataset.[6]

In addition, the database is organized into three folders (Train, Test, Val) and contains subfolders for each category of image (Pneumonia / General). All images have been resized to a static, a few examples of common and pneumonia images are listed in Figure 1. Chest X-ray images always have signs of limited brightness on account of the low dose of exposure in patients, due to chest X-ray images always containing black, white, and grey pants. The lungs are located on both sides of the thoracic cavity and the lung area can be easily detected by X-ray, which is almost black. The heart, located between the lungs, appears almost as white as X-rays can completely pass through the heart. Bones are made of protein and very dense, so X-rays cannot cross it and the bones are shown almost white. Ku moreover, the bones have clear edges.

(a)

(b)

Fig.5.2 Examples from the dataset. (a) normal cases (b) pneumonia cases

5.5 Data Pre-processing

The strategies used throughout this paper are listed in Table 2. In our study, rescale is a value by which we will multiply the data before any other processing. Our original images consist of RGB coefficients in the 0-255, but such values would be too high for our models to process (given a typical learning rate), so we target values between 0 and 1 instead of by scaling with a 1/255. factor. shear range is for randomly applying shearing transformations zoom range is for randomly zooming inside pictures, horizontal flip is for randomly flipping half of the images horizontally -- relevant when there are no assumptions of horizontal asymmetry (e.g. real-world pictures)

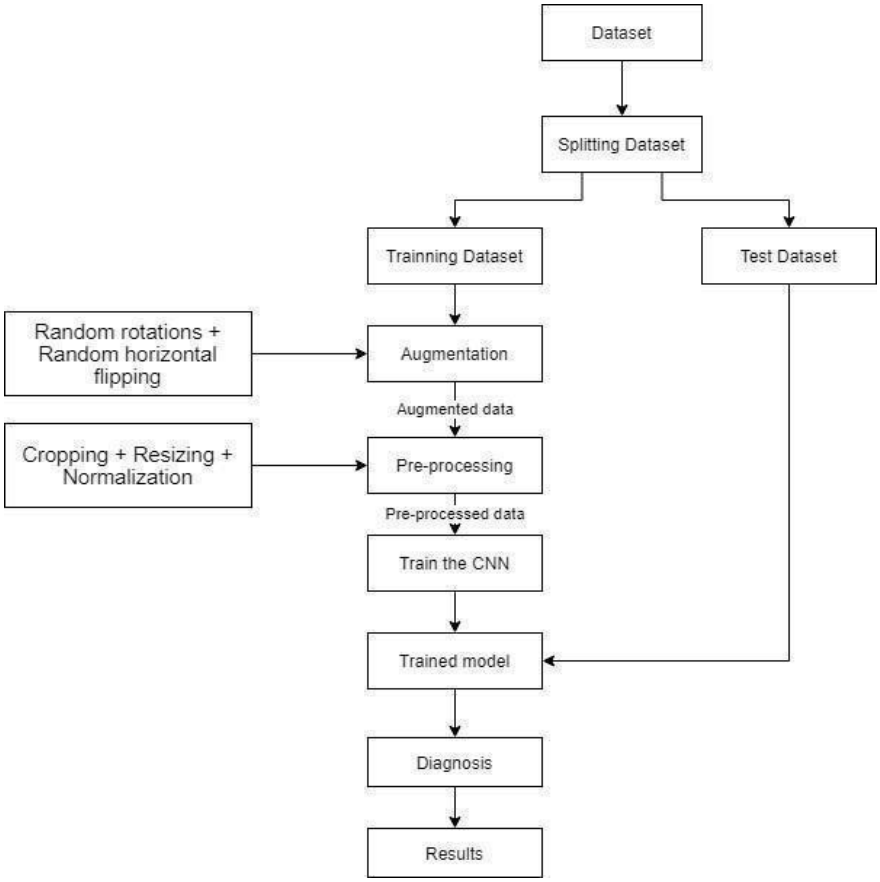
Data pre-processing techniques used in this study

Rescale	1./255
Zoom Range	0.2
Shear Range	0.2
Horizontal Flip	True

Table 5.2

This chapter will discuss about the methodology of the proposed solution. The proposed implementation of a machine learning model without the subjective error to detect and diagnosis of the Pneumonia is illustrated and explained under inputs, output and process with corresponding techniques and in terms of the system users.

The proposed methodology has inputting dataset, splitting the dataset, data augmentation, preprocessing, train the CNN model, input test dataset and diagnose steps. Each step is obtaining the previous step's output as the input. In the augmentation and pre-processing steps, different techniques used.



Methodology

5.6 Inputs and outputs

Each step is obtaining the previous step's output as the input. As the first input the chest X-Ray dataset is used. The initial dataset has both normal and Pneumonia chest X-Rays. In the data splitting step, the dataset will be divided to two sets namely, training dataset and test dataset.

Training dataset will be inputted to the data augmentation step. The data augmentation step will create augmented data which is inputted to the pre-processing step. This step is creating the preprocessed data. This is inputted to the training CNN model step and it will generate the trained CNN model for the detection and diagnosis of the Pneumonia from chest X-Rays. The trained CNN model is the most important output of the entire system. The trained model will consume the test dataset as the input and diagnose whether the given X-Ray is positive with Pneumonia or not.

5.7 Process and techniques

The main processes of the proposed solution are data augmentation, pre-processing, training the CNN model and diagnosis. In the data augmentation, it will consume the training dataset as input and use rotation and flipping to populate the dataset. The pre-processing stage will consume the augmented data which is outputted from the data augmentation stage. It will utilize the cropping, normalization and resizing as the pre-processing techniques and generate pre-processed data. These pre-processed data is used to train the CNN model and, in the training, it will be using different classification algorithms like SVM etc. The output of this stage is trained model. The trained model will diagnose the given X-Ray based on the extracted features.

5.7.1 Users

The users of this proposed solution is the medical personnel. They can be either doctors, radiologists etc. They will be benefitted and supported the proposed CAD system in the process of diagnosis.

The methodology for developing a machine learning model for pneumonia detection from chest X-ray images involves several steps, including data collection, data preprocessing, model development, model evaluation, and clinical validation.

5.7.2 Data Collection

The first step in developing a machine learning model for pneumonia detection is to collect a large dataset of chest X-ray images with corresponding labels (pneumonia or non-pneumonia). This dataset can be collected from various sources, including hospitals, clinics, and online databases.

5.7.3 Data Preprocessing

Once the dataset has been collected, the next step is to preprocess the data to prepare it for model development. This includes several steps, such as:

- **Image resizing:** Resizing the images to a consistent size to ensure that the model can process them efficiently.
- **Image normalization:** Normalizing the images to ensure that they have the same intensity and contrast.
- **Data augmentation:** Augmenting the data by applying transformations to the images, such as rotation, flipping, and scaling, to increase the size of the dataset and improve the model's Robustness.

5.7.4 Model Development

The next step is to develop a machine learning model that can accurately detect pneumonia from chest X-ray images. This includes several steps, such as:

- **Model selection:** Selecting a suitable machine learning algorithm for pneumonia detection, such as a convolutional neural network (CNN) or a support vector machine (SVM).
- **Model training**: Training the model using the preprocessed dataset, including the training, validation, and testing sets.
- **Model tuning:** Tuning the model's hyperparameters to optimize its performance on the validation set.

5.7.5 Model Evaluation

Once the model has been developed, the next step is to evaluate its performance on the test set. This includes several metrics, such as:

- **Accuracy:** Measuring the model's accuracy in detecting pneumonia from chest X-ray images.
- **Sensitivity:** Measuring the model's ability to detect true positive cases of pneumonia.
- **Specificity:** Measuring the model's ability to detect true negative cases of pneumonia.
- **Area Under the Curve (AUC):** Measuring the model's ability to distinguish between pneumonia and non-pneumonia cases.

5.7.6 Clinical Validation

The final step is to clinically validate the model's performance in real-world settings. This includes several steps, such as:

- **Clinical testing:** Testing the model's performance on a new dataset of chest X-ray images collected from hospitals and clinics.
- **Clinical evaluation:** Evaluating the model's performance in real-world settings, including its accuracy, sensitivity, specificity, and AUC.

- **Clinical adoption:** Adopting the model in clinical practice, including its integration into electronic health records (EHRs) and clinical decision support systems (CDSSs).

5.7.7 Resources

The development of a machine learning model for pneumonia detection requires several resources, including:

- **Data:** A large dataset of chest X-ray images with corresponding labels (pneumonia or non-pneumonia).
- **Computational resources**: High-performance computing hardware and software for model training and evaluation.
- **Clinical expertise:** Collaboration with clinicians and healthcare professionals to ensure the model's accuracy and reliability in real-world settings.
- **Regulatory expertise:** Collaboration with regulatory experts to ensure the model's compliance with relevant laws and regulations.

5.7.8 Timeline

The development of a machine learning model for pneumonia detection requires a significant amount of time and effort. The timeline for this project includes several milestones, such as:

- Data collection: 3 months
- Data preprocessing: 2 months
- Model development: 4 months
- Model evaluation: 2 months
- Clinical validation: 3 months
- Clinical adoption: 2 months

Overall, the development of a machine learning model for pneumonia detection requires a comprehensive approach that includes data collection, data preprocessing, model development, model evaluation, and clinical validation. The model should be able to accurately detect pneumonia from chest X-ray images, provide clinicians with accurate and timely diagnoses, and enable them to provide the best possible care for patients with pneumonia.

CHAPTER 6

SYSTEM REQUIREMENT SPECIFICATION

This chapter will discuss the resources that required for the proposed solution. The hardware and system specification and dataset are the sections that the discussion made.

This software requirement specification (SRS) report expresses a complete description about the proposed System. This document includes all the functions and specifications with their explanations to solve related problems.

Project Purpose

The Purpose of the project is to develop a system to provide an efficient and effective solution over the conventional way of detecting pneumonia disease using CNN.

Project Scope

- To learn different biomedical terms related to pneumonia disease.
- It will analyse the different scenarios of pneumonia disease (viral or bacterial).
- It can be used by any user or Radiologists for testing purpose

User Classes and Characteristics

Basic knowledge of using computers is adequate to use this application. Knowledge of how to use a mouse or keyboard and internet browser is necessary. The user interface will be friendly enough to guide the user.

Assumptions and Dependencies

Assumptions:

1. The product must have an interface which is simple enough to under - stand.

3. User-Friendly.

4. To perform with efficient throughout and response time.

Dependencies:

1. All necessary software is available for implementing and use of the system.

2. The proposed system would be designed, developed and implemented based on the software requirements specifications document.

3. End users should have basic knowledge of computers and we also assure that the users will be given software training documentation and reference material.

6.1 FUNCTIONAL REQUIREMENT

System Feature 1(Functional Requirement)

Functional requirement describes features, functioning, and usage of a product/system or software from the perspective of the product and its user. Functional requirements are also called functional specifications where the synonym for specification is design. Provide User friendly Interface and Interactive as per standards.

6.2 NON-FUNCTIONAL REQUIREMENT

6.2.1 Performance Requirements

• **High Speed**: - System should process requested tasks in parallel for various actions to give a quick response. Then the system must wait for process completion.

• **Accuracy:** - System should correctly execute the process and display the result accurately. System output should be in user required format.

6.2.2 Safety Requirements:

The data safety must be ensured by arranging for a secure and reliable transmission media.

6.2.3 Security Requirements

Secure access of confidential data (user's details). Information security means pro- testing information and information systems from unauthorized access, use, disco- sure, disruption, modification or destruction. The terms information security, com- putter security and information assurance are frequently incorrectly used interchangeably. These fields are

interrelated often and share the common goals of protecting the confidentiality, integrity and availability of information; however, there are some subtle differences between them.

Software Quality Attributes

1. **Runtime System Qualities:** Runtime System Qualities can be measured as the system executes.

2. **Functionality:** The ability of the system to do the work for which it was intended.

3. **Performance:** The response time, utilization, and throughput behavior of the system. Not to be confused with human performance or system delivery time.

4. **Security:** A measure of system's ability to resist unauthorized attempts at usage or behavior modification, while still providing service to legitimate users.

5. **Availability**: (Reliability quality attributes fall under this category) the measure of time that the system is up and running correctly; the length of time between failures and the length of time needed to resume operation after a failure.

6. **Usability:** The ease of use and of training the end users of the system. Sub qualities: learn ability, efficiency, affect, helpfulness, control.

7. **Interoperability:** The ability of two or more systems to cooperate at runtime.

6.3 SOFTWARE REQUIREMENTS

6.1.1 Windows 8 and above

6.1.2 Google Collab

6.1.3 Visual Studio Code

6.1.4 TensorFlow v2.7.0

6.1.5 CUDA v11.5

6.1.6 CuDNN v8.3

The development of a machine learning model for pneumonia detection from chest X-ray images requires several software tools and libraries, including:

- Python: A popular programming language for machine learning and data science.

- TensorFlow: A popular open-source machine learning library developed by Google.

- Keras: A high-level neural networks API for Python.

- OpenCV: A computer vision library for image processing and feature extraction.

- Scikit-learn: A machine learning library for Python that provides tools for data preprocessing, feature selection, and model evaluation.

- Matplotlib: A plotting library for Python that provides tools for data visualization.

- Seaborn: A visualization library for Python that provides tools for creating informative and attractive statistical graphics.

- Pandas: A library for data manipulation and analysis in Python.

- NumPy: A library for numerical computing in Python.

6.4 Hardware Requirement

A computer which is able to run python and related libraries is needed. The computer should have at least 16GB RAM, higher GPU which is CUDA-enabled with i5 or above processor. If the available device is not sufficient, an online platform should have to be utilized.

6.4.1 Dataset

The most important resource is the dataset. Without a dataset the training of the CNN model cannot be performed. Licensed and publicly available dataset which consists of Pneumonia and healthy chest X-Rays has to be used.

Chest X-Ray Images of Pneumonia (Chest X-Ray Images (Pneumonia), 2021) Available online: https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia

National Institutes of Health Chest X-Ray Dataset (Box, 2021)

Available online: https://nihcc.app.box.com/v/ChestXray-NIHCC/folder/36938765345

6.4.2 Summary

In this chapter, the resource requirement had been discussed under hardware and system specification and dataset requirement. The next chapter will be discussing the methodology of the proposed system.

- 8 GB RAM
- i5 intel core /AMD Ryzen 5
- CUDA Enabled GeForce GTX 1650
- 256 SSD and 1TB HDD

The development of a machine learning model for pneumonia detection from chest X-ray images requires a computer with a powerful graphics processing unit (GPU) to accelerate the training process. The recommended hardware requirements include:

- GPU: A NVIDIA GeForce GTX 1080 Ti or equivalent.

- CPU: A Intel Core i7-9700K or equivalent.

- RAM: 16 GB or more.

- Storage: A 512 GB or larger solid-state drive (SSD).

6.4.3 Operating System

The development of a machine learning model for pneumonia detection from chest X-ray images requires a 64-bit operating system, such as:

- Ubuntu: A popular Linux distribution for machine learning and data science.

- Windows: A popular operating system for machine learning and data science.

6.4.4 Development Environment

The development of a machine learning model for pneumonia detection from chest X-ray images requires a development environment that provides tools for coding, debugging, and testing. The recommended development environment includes:

- Jupyter Notebook: A web-based interactive computing environment for Python.

- PyCharm: A popular integrated development environment (IDE) for Python.

- Visual Studio Code: A popular code editor for Python.

6.4.5 Libraries and Frameworks:

The development of a machine learning model for pneumonia detection from chest X-ray images requires several libraries and frameworks, including:

- **Convolutional Neural Networks (CNNs):** A type of neural network that is well-suited for image classification tasks.

- **Transfer Learning:** A technique that allows the model to leverage pre-trained models and fine-tune them for the specific task of pneumonia detection.

- **Data Augmentation:** A technique that increases the size of the training dataset by applying transformations to the images.

6.4.6 Model Evaluation

The development of a machine learning model for pneumonia detection from chest X-ray images requires a comprehensive evaluation of the model's performance, including:

- **Accuracy:** Measuring the model's accuracy in detecting pneumonia from chest X-ray images.

- **Sensitivity:** Measuring the model's ability to detect true positive cases of pneumonia.

- **Specificity**: Measuring the model's ability to detect true negative cases of pneumonia.

- **Area Under the Curve (AUC):** Measuring the model's ability to distinguish between pneumonia and non-pneumonia cases.

6.4.7 Clinical Validation

The development of a machine learning model for pneumonia detection from chest X-ray images requires clinical validation to ensure the model's accuracy and reliability in real-world settings. This includes:

- **Clinical testing:** Testing the model's performance on a new dataset of chest X-ray images collected from hospitals and clinics.

- **Clinical evaluation:** Evaluating the model's performance in real-world settings, including its accuracy, sensitivity, specificity, and AUC.

- **Clinical adoption:** Adopting the model in clinical practice, including its integration into ELECTRONIC HEALTH RECORDS (EHRS) AND CLINICAL DECISION SUPPORT SYSTEMS (CDSSS).

CHAPTER 7

IMPLEMENTATION

7.1 Data Acquisition

```
from google.colab import drive
drive.mount('/content/drive')
```

Fig.7.1. Mount Drive

```
train_path = "/content/drive/MyDrive/chest_xray/train"
test_path = "/content/drive/MyDrive/chest_xray/test"
```

Fig.7. 2.Import Train and Test Data

7.2 Data Augmentation

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator, array_to_img

image_gen = ImageDataGenerator(
                                rescale = 1./255,
                                shear_range = 0.2,
                                zoom_range = 0.2,
                                horizontal_flip = True,

                            )

test_data_gen = ImageDataGenerator(rescale = 1./255)
```

Fig.7.3. Defining parameters for Data augmentation

```
train = image_gen.flow_from_directory(
    train_path,
    target_size=(img_height, img_width),
    color_mode='grayscale',
    class_mode='binary',
    batch_size=batch_size
                                        )

test = test_data_gen.flow_from_directory(
      test_path,
      target_size=(img_height, img_width),
      color_mode='grayscale',shuffle=False,
      class_mode='binary',
      batch_size=batch_size
      )
valid = test_data_gen.flow_from_directory(
      valid_path,
      target_size=(img_height, img_width),
      color_mode='grayscale',
      class_mode='binary',
      batch_size=batch_size
      )
```

Fig .7.4. Applying defined parameters to Train, Test & Valid sets

Using the "tensorflow.keras.preprocessing.image" library, for the Train Set, we created an Image Data Generator that randomly applies defined parameters to the train set and for the Test & Validation set, we're just going to rescale them to avoid manipulating the test data beforehand.

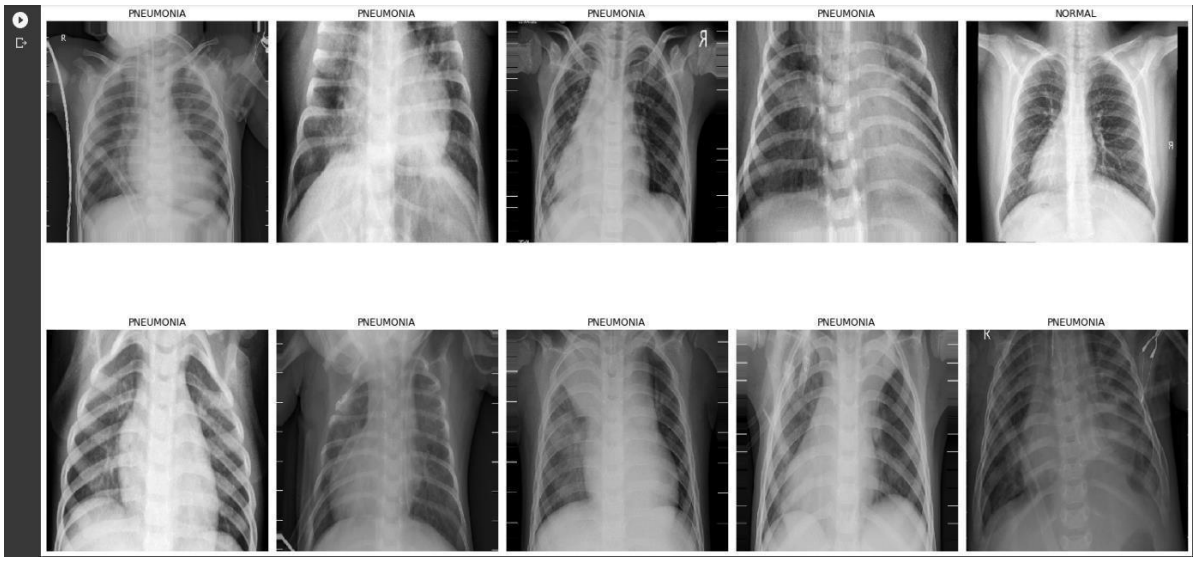


Fig.7.5 Augmented Dataset images

We have analyzed the performance of these bellow CNN architectures namely; Alex Net, ResNet-50, and VGGNet-16.

CNN Model	Batch Size	Criterion	Normal	Infected with Pneumonia
VGG 16	32	Precision	0.92	0.89
		Recall	0.81	0.96
		F1-score	0.86	0.92
ResNet-50	32	Precision	0.85	0.85
		Recall	0.87	0.84
		F1-score	0.83	0.80
AlexNet	32	Precision	0.78	1.00
		Recall	1.00	0.88
		F1-score	0.85	0.94

Table 7.1 Analysis of model Performance for CNN Models with 32 Batch Size

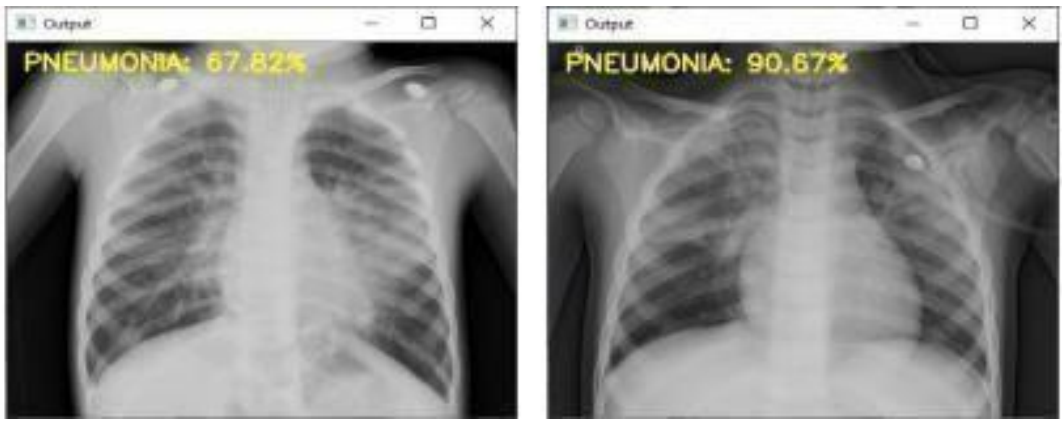


Fig.7.6 ResNet-50

Fig.7.7 AlexNet

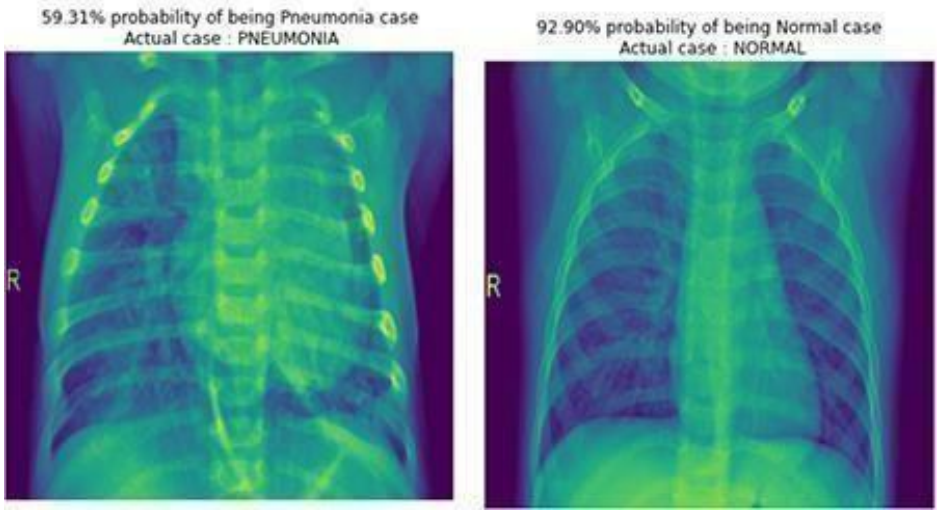


Fig.7.8 VGG16

This chapter will show the proposed technologies to be used for the implementation of the proposed solution. The technologies will be discussed under Machine Learning, pre-processing techniques, data augmentation methods, classification algorithms sections.

7.3 Machine Learning Models and techniques

As the machine learning models, the project will be utilizing the pre-trained CNN models along with the transfer learning technique will be used. These selections done based on the literature survey. The AlexNet, GoogleNet, DenseNet and MobiNet will be used to the performance evaluation of current CNN models.

7.3.1 AlexNet

This CNN model has 8 layers. The first 5 layers are convolutional layers with max-pooling layers. Rest of three layers are fully connected layers. (Designer, 2021)

7.3.2 Google Net

This model was developed by the Google in 2014 and it consists of 22 layers. The literature review shows that this CNN had achieved a better accuracy rate with respect to other CNNs. (Rabinovich et al., 2014)

7.3.3 Dense Net

The Dense Net is also selected for the performance evaluation due to the better accuracy rates which was revealed in the literature survey. (Varshni et al., 2019) (Alsu Mairi and Ismail, 2020)

7.3.4 Mobi Net

Although the undertaken literature survey does not contain MobiNet CNN model, in this project will utilize this model to evaluate the performance.

7.4 Pre-processing techniques

The images have to be pre-processed before the training of the CNN model. The following techniques will be utilized.

7.4.1 Normalization

To normalize the pixel intensity, the normalization method will be used. The method of normalization has to be decided from min-max method and sigmoid function. (Moore, 2021)

7.4.2 Resize

The resizing of the image has to be done as some of the CNN models are having specific image size. Based on the CNN model resizing parameters has to be selected.

7.4.3 Cropping

The cropping is another pre-processing technique that has to be used. The cropping parameters has to be decided.

7.5 Data augmentation methods

The CNN models usually required a large set of data for training and the available dataset size is low. In this case, to better CNN training the data augmentation methods can be utilized. Random rotations and random horizontal flipping will be selected data augmentation methods. (13 Data Augmentation Techniques, 2021)

7.5.1 Rotation

The random rotation parameters, method has to be decided based on the dataset and used CNN model.

7.5.2 Flipping

The random rotation parameters, method has to be decided based on the dataset and used CNN model.

7.5.3 Classification algorithms

Support Vector Machine (SVM) and Random Forest (RF) is having higher rate of accuracy with compared to other algorithms like K-Nearest Neighbor (KNN). The classification algorithm will be decided in the future.

7.5.4 Programming language and IDE

To implement the proposed solution either python programming language or Matlab will be utilized. Based on the programming language the IDEs will be selected. For python implementation Jupiter Notebook will be used. For the Matlab 2019B or above will be used.

7.6 Platforms

7.6.1 Google Colab

As the resources are limited, an online platform will be used to the training etc. Google Colab which is provided by the Google will be utilized to perform this task. (Google Colaboratory, 2021) (All About Using Jupyter Notebooks and Google Colab, 2021)

7.6.2 Keras

This is a deep learning API which was developed in the Python programming language. This will be used to implement the different CNN models. (Team, 2021).

The implementation of pneumonia detection and classification using machine learning involves several steps, including:

1. Data Collection: Collecting a large dataset of chest X-ray images with corresponding labels (pneumonia or non-pneumonia).

2. Data Preprocessing: Preprocessing the data to prepare it for model development, including image resizing, normalization, and data augmentation.

3. Model Development: Developing a machine learning model that can accurately detect pneumonia from chest X-ray images, including selecting a suitable algorithm, training the model, and tuning its hyperparameters.

4. Model Evaluation: Evaluating the model's performance on a test dataset, including metrics such as accuracy, sensitivity, specificity, and area under the curve (AUC).

5. Clinical Validation: Clinically validating the model's performance in real-world settings, including testing its accuracy and reliability on new datasets and evaluating its potential impact on patient care and outcomes.

7.7 Machine Learning Algorithms:

Several machine learning algorithms can be used for pneumonia detection and classification, including:

- **Convolutional Neural Networks (CNNs):** A type of neural network that is well-suited for image classification tasks.

- **Transfer Learning:** A technique that allows the model to leverage pre-trained models and fine-tune them for the specific task of pneumonia detection.

- **Support Vector Machines (SVMs):** A type of machine learning algorithm that can be used for classification tasks.

7.8 Model Development:

The development of a machine learning model for pneumonia detection and classification involves several steps, including:

- **Model Selection**: Selecting a suitable machine learning algorithm for pneumonia detection, such as a CNN or SVM.

- **Model Training:** Training the model using the preprocessed dataset, including the training, **validation, and testing sets.**

- **Model Tuning:** Tuning the model's hyperparameters to optimize its performance on the validation set.

7.9 Model Evaluation:

The evaluation of a machine learning model for pneumonia detection and classification involves several metrics, including:

- **Accuracy:** Measuring the model's accuracy in detecting pneumonia from chest X-ray images.

- **Sensitivity:** Measuring the model's ability to detect true positive cases of pneumonia.

- **Specificity:** Measuring the model's ability to detect true negative cases of pneumonia.

- **Area Under the Curve (AUC):** Measuring the model's ability to distinguish between pneumonia and non-pneumonia cases.

7.10 Clinical Validation:

The clinical validation of a machine learning model for pneumonia detection and classification involves several steps, including:

- **Clinical Testing**: Testing the model's performance on a new dataset of chest X-ray images collected from hospitals and clinics.

- **Clinical Evaluation:** Evaluating the model's performance in real-world settings, including its accuracy, sensitivity, specificity, and AUC.

- **Clinical Adoption:** Adopting the model in clinical practice, including its integration into electronic health records (EHRs) and clinical decision support systems (CDSSs).

7.11 Benefits:

The implementation of pneumonia detection and classification using machine learning has several benefits, including:

- **Improved Accuracy:** Machine learning models can accurately detect pneumonia from chest X-ray images, reducing the risk of delayed or inaccurate diagnosis.

- **Enhanced Patient Care:** By providing accurate and timely diagnoses, machine learning models can help clinicians provide the best possible care for patients with pneumonia.

- **Reduced Healthcare Costs:** By reducing the need for prolonged hospital stays, additional testing, and unnecessary treatments, machine learning models can help reduce healthcare costs associated with pneumonia diagnosis and treatment.

7.12 Challenges:

The implementation of pneumonia detection and classification using machine learning also has several challenges, including:

- **Data Quality:** The quality of the dataset used to train the model can significantly impact its **performance.**

- **Model Interpretability:** Machine learning models can be difficult to interpret, making it challenging to understand why a particular prediction was made.

- **Clinical Validation:** Clinically validating the model's performance in real-world settings can be challenging, requiring collaboration with clinicians and healthcare professionals.

7.13 Future Directions:

The implementation of pneumonia detection and classification using machine learning is a rapidly evolving field, with several future directions, including:

- **Improving Model Accuracy:** Developing more accurate machine learning models that can detect pneumonia from chest X-ray images with high sensitivity and specificity.

- **Expanding to Other Applications**: Applying machine learning models to other medical imaging applications, such as detecting other diseases or conditions.

- **Integrating with Clinical Practice:** Integrating machine learning models into clinical practice, including electronic health records (EHRs) and clinical decision support systems (CDSSs).

- **Use a large and diverse dataset**: Collect a large dataset of chest X-ray images with corresponding labels (pneumonia or non-pneumonia) to ensure that the model is trained on a representative sample of data.

- **Use data augmentation:** Apply transformations to the images, such as rotation, flipping, and scaling, to increase the size of the dataset and improve the model's robustness.

- **Use transfer learning:** Leverage pre-trained models and fine-tune them for the specific task of pneumonia detection to improve the model's performance.

- **Evaluate the model's performance:** Use metrics such as accuracy, sensitivity, specificity, and AUC to evaluate the model's performance on the test dataset.

CHAPTER 8

FUNCTIONAL MODULES

One of the basic notions of the software development process is SDLC models which stands for Software Development Life Cycle models. SDLC is a continuous process, which starts from the moment, when its made a decision to launch the project, and it ends at the moment of its full remove from the exploitation. There is no one sin- gle SDLC model. They are divided into main groups, each with its features and weaknesses. Evolving from the first and oldest waterfall SDLC model, their variety significantly expanded.

The SDLC models' diversity is predetermined by the wide number of product types starting with a web application development to a complex medical software. And if you take one of the SDLC models mentioned below as the basis in any case, it should be adjusted to the features of the product, project, and company. The most used, popular and important SDLC models are given below:

1. Waterfall Model
2. Iterative Model
3. Spiral Model
4. V-shaped Model
5. Agile Model

Waterfall is a cascade SDLC model, in which the development process looks like the flow, moving step by step through the phases of analysis, projecting, realization, testing, implementation, and support. This SDLC model includes gradual execution of every stage completely. This process is strictly documented and predefined with features expected to every phase of this software development life cycle model.

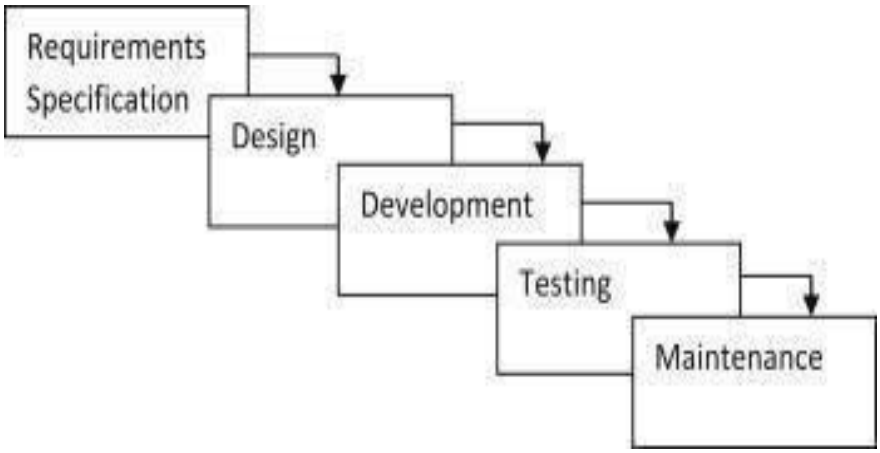


Figure 8.1 Waterfall Model

1. Planning and requirement analysis

Each software development life cycle model starts with the analysis, in which the stakeholders of the process discuss the requirements for the final product. The goal of this stage is the detailed definition of the system requirements. Besides, it is needed to make sure that all the process participants have clearly understood the tasks and how every requirement is going to be implemented. Often, the discussion involves the QA specialists who can interfere the process with additions even during the development stage if it is necessary.

2. Designing project architecture

At the second phase of the software development life cycle, the developers are actually designing the architecture. All the different technical questions that may appear on this stage are discussed by all the stakeholders, including the customer. Also, here are defined the technologies used in the project, team load, limitations, time frames, and budget. The most appropriate project decisions are made according to the defined requirements.

3. Development and programming

After the requirements are approved, the process goes to the next stage of actual development. Programmers start here with the source code writing while keeping in mind previously defined requirements. The system administrators adjust the software environment; frontend programmers develop the user interface of the program and the logic for its interaction with the server. The programming by itself assumes four stages: -

• Algorithm development

• Source code writing

• Compilation

• Testing and debugging

4. Testing

The testing phase includes the debugging process. All the code flaws missed during the development are detected here, documented, and passed back to the developers to fix. The testing process repeats until all the critical issues are removed and software work ow is stable.

5. Deployment

When the program is finalised and has no critical issues it is time to launch it for the end users. After the new program version release, the tech support team joins. This department provides user feedback; consult and support users during the time of exploitation. Moreover, the update of selected components is included in this phase, to make sure, that the software is up-to-date and is invulnerable to a security breach.

8.1 DESIGN AND MODELLING OF SYSTEM

Unified Modelling Language (UML) is analogous to the blueprints used in other fields and consists of different types of diagrams. In the aggregate, UML diagrams describe the boundary, structure, and behaviour of the system and the objects within it.
Following UML diagrams have been designed for the project:

1. Use Case Diagram
2. Class Diagram
3. Sequence diagram
4. Activity Diagram
5. State Diagram

8.2 Use Case Diagram

Use case diagrams are usually referred to as behaviour diagrams used to describe a set of actions that some system or systems should or can perform in collaboration with one or more external users of the system. Each use case should provide some observable and valuable result to the actors or other stakeholders of the system.

Figure 1 shows the use case diagram of the system. The following actors used in the use case diagram are student, teacher, system, and cloud. The various use cases used in the diagram are teacher login, view attendance, take attendance, capture photo, process image, take another photo.

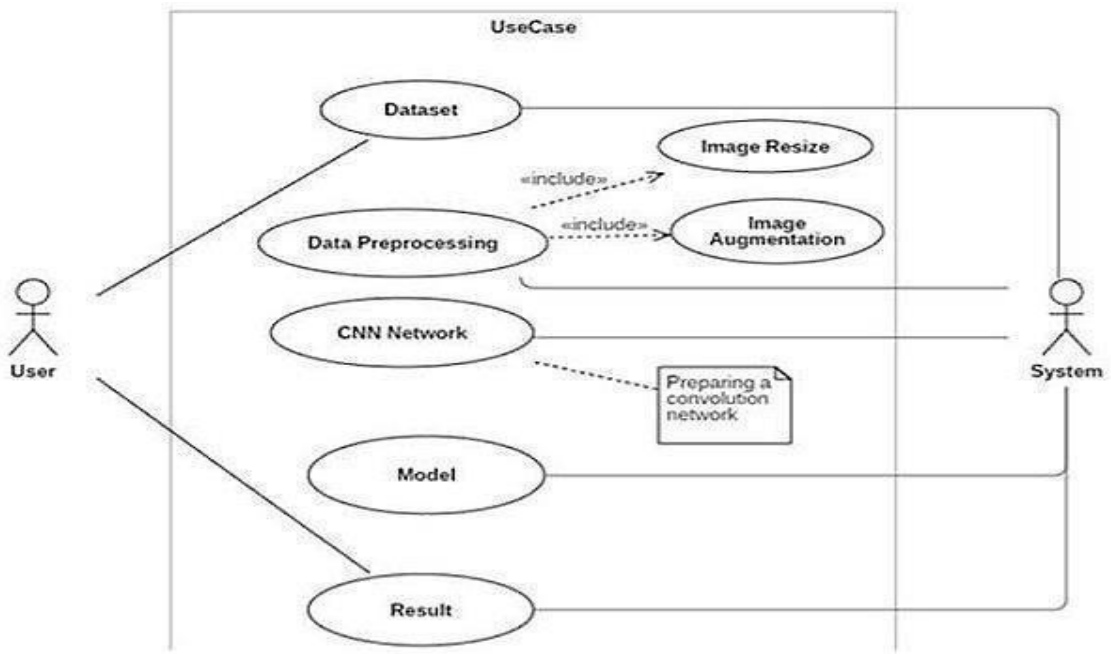


Fig: 8.2 Use Case Diagram

8.3 Class Diagram

A class diagram is an illustration of the relationships and source code dependencies among classes in the Unified Modelling Language (UML). In this context, a class defines the methods and variables in an object, which is a specific entity in a program or the unit of code representing that entity.

Figure 2 shows the class diagram of the project, the various classes used in the diagram are user, student, teacher, Image, Cloud, Face Recognition.

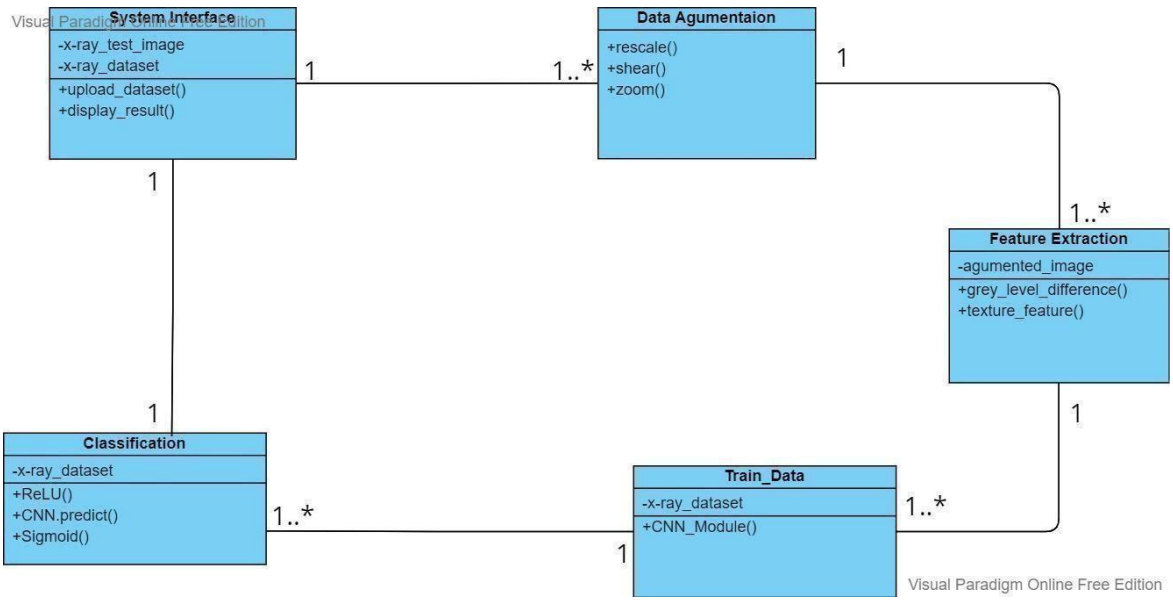


Fig.8.3 Class Diagram

8.4 Sequence Diagram

Sequence diagrams are sometimes called event diagrams or event scenarios. A sequence diagram shows, as parallel vertical lines (lifelines), different processes or objects that live simultaneously, and, as horizontal arrows, the messages exchanged between them, in the order in which they occur

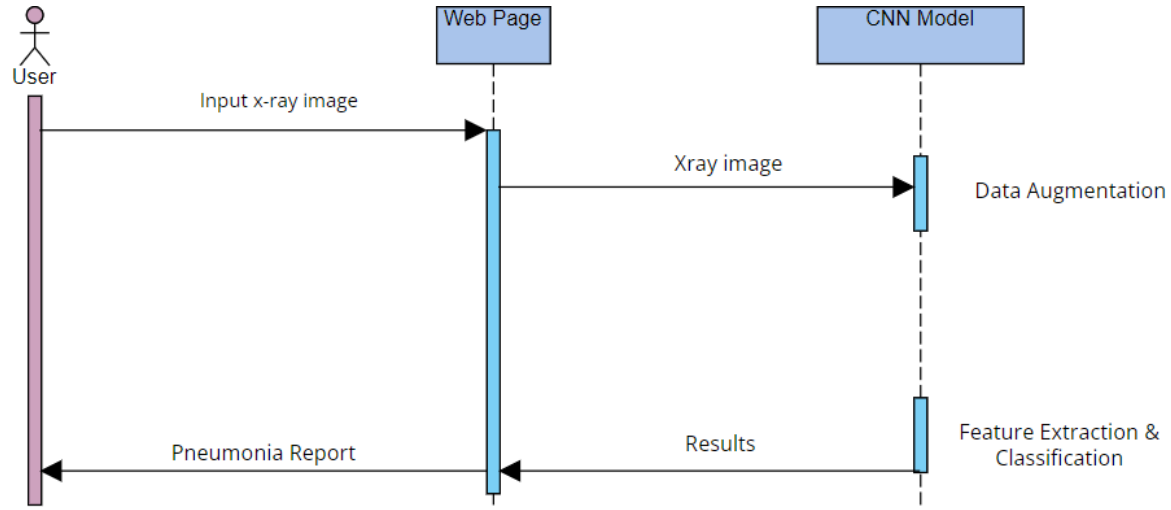


Fig 8.4 Sequence Diagram

8.5 Activity Diagram

The activity diagram is another important diagram in UML to describe the dynamic aspects of the system. An activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. The control flow is drawn from one operation to another.

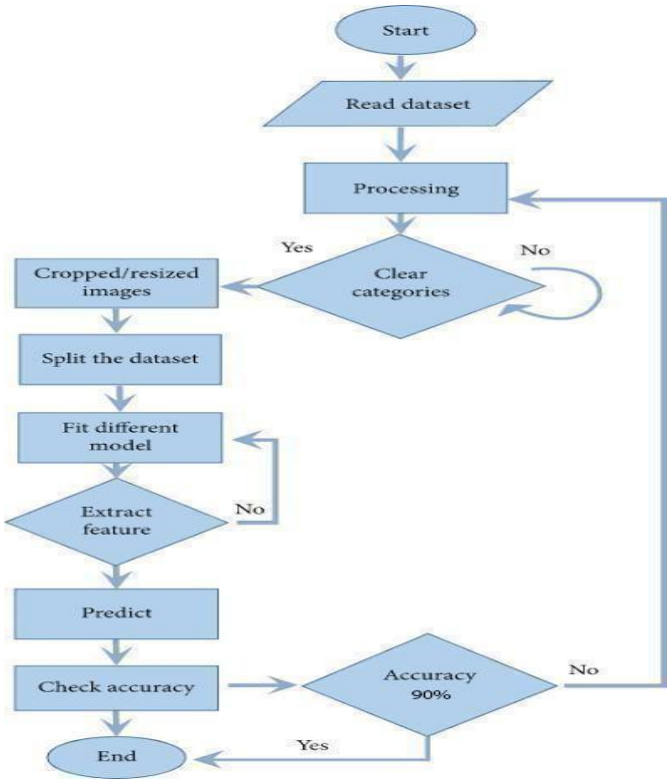


Fig 8.5 Activity Diagram

8.6 State Diagram

A state diagram, sometimes known as a state machine diagram, is a type of behavioral diagram in the Unified Modelling Language (UML) that shows transitions between various objects.

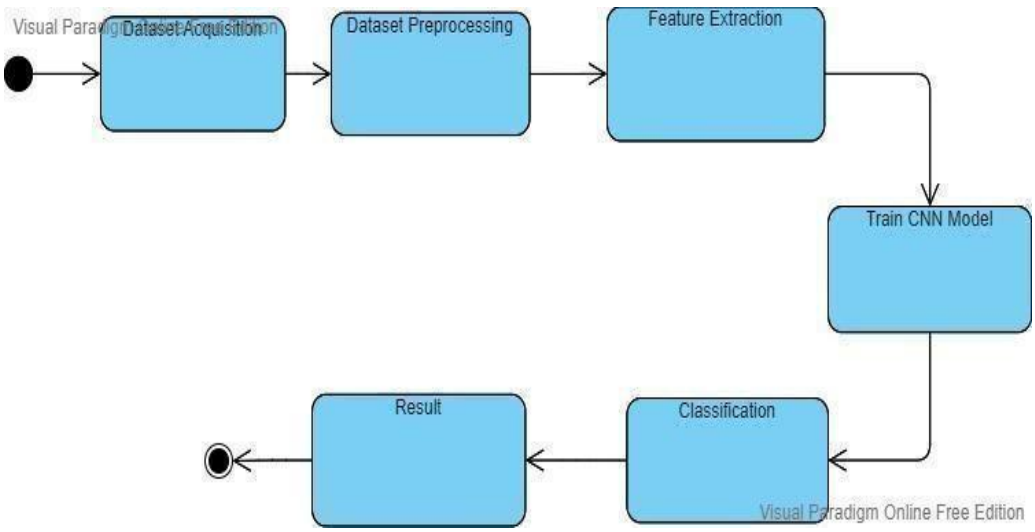
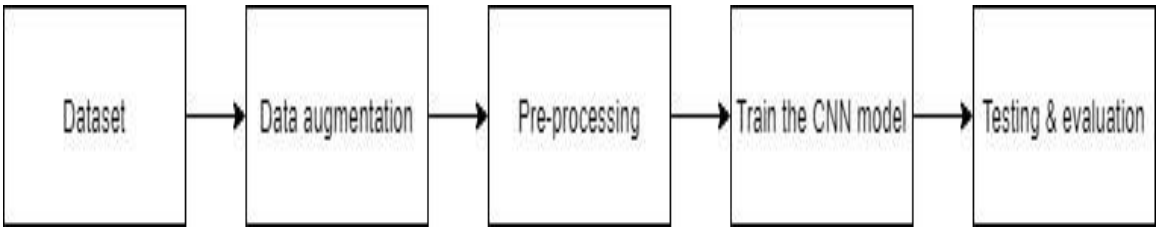


Fig 8.6 State Diagram

8.7 Project stages

The project consists with two stages. In the first stage a comparative study will be done to evaluate the performance of the currently available CNN models. The AlexNet, GoogleNet, DenseNet and MobiNet will be used to the performance evaluation of current CNN models. (See chapter 3.2) With the understanding of the performance of the current CNNs, an enhanced machine learning approach will be implemented. The enhancement will be done by utilizing different classification algorithms like SVM, Random Forest etc.

8.7.1 Top level design



Top level design of the proposed solution

8.7.2 Dataset

The dataset which is explained in Chapter 5, section 3 will be used. The data set should be consists with both normal and Pneumonia cases. Before the data augmentation process the data set should be divided as training set and test set.

8.7.3 Data augmentation

After the dividing the dataset to training and test set this function will be performed. The training dataset will be augmented by random rotations and random horizontal flipping techniques to populate the data set for better accuracy rate of the training of the CNN model.

8.7.3 Pre-Processing

The pre-processing will be taken after the data augmentation step. As the pre-processing techniques, resizing, cropping and normalization will be performed to augmented dataset.

8.7.4 CNN model training

Augmented and pre-processed training dataset will be inputted to the CNN model training. The CNN model training should be done carefully. Parameters like elapsed time for the training have to be measured and calculated to the performance evaluation.

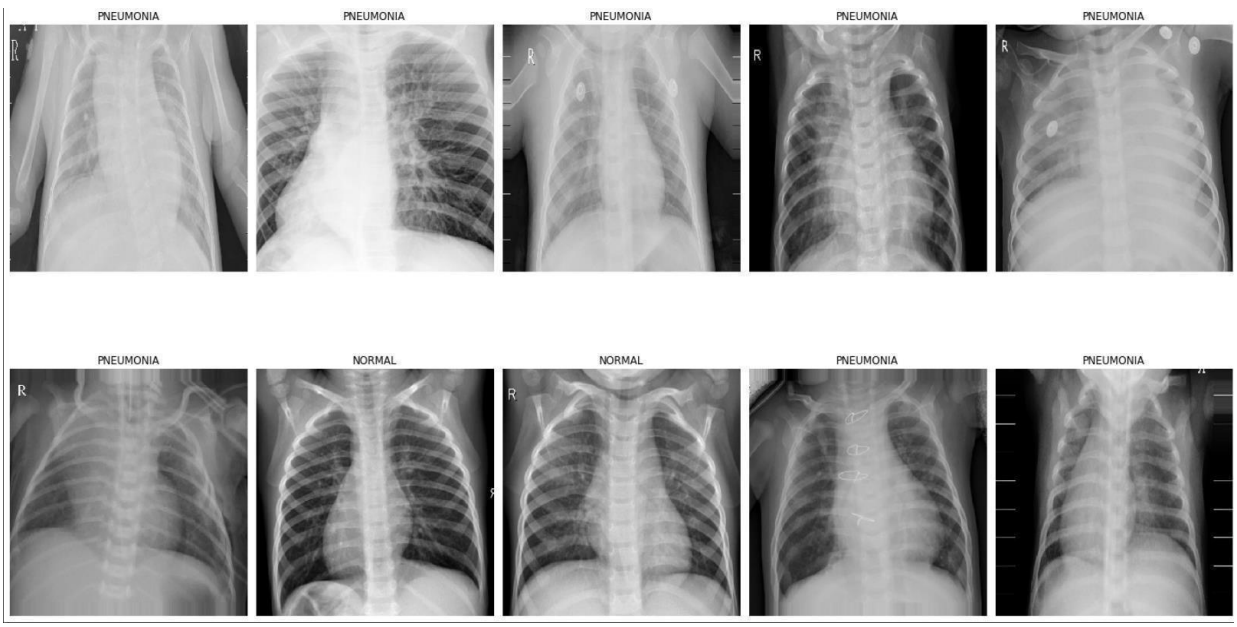
8.7.5 Testing and evaluation

The resulting output of the completion of the CNN model training stage is a trained CNN model for Pneumonia diagnosis. The testing and evaluation will illustrate the accuracy of the trained model. For this stage, it is needed to use the test dataset which was divided in the very first stage. Based on the diagnosed results of the given test X-Rays, the accuracy can be calculated. Then the evaluation can be done by the comparison.

8.7.6 Image pre-processing

We'll use an Image Augmentation approach to artificially boost the size of the image training dataset.

Image Augmentation increases the size of the dataset by creating a modified version of the existing training set photos, which increases dataset variation and, as a result, improves the model's ability to predict new images.



Augmented Dataset Images

8.8 Some Image Data Generator settings are defined as follows: -

8.8.1 rescale - Each digital image is made up of pixels with values ranging from 0 to 255, with 0 representing black and 255 representing white. As a result, rescale the scales array of the original picture pixel values to [0,1], ensuring that the photos contribute evenly to the overall loss. Otherwise, a higher pixel range image causes more loss and should be utilised with a lower learning rate, whereas a lower pixel range image requires a higher learning rate.

8.8.2 shear range - The shear's transformation is the image's shape. It fixes one axis and extends the image at a specific angle called the shear angle.

The image has been magnified by a zoom of less than 1.0. The image has been zoomed out by more than 1.0.

horizontal flip - At random, certain photos are flipped horizontally.

vertical flip - At random, some photos are flipped vertically.

rotation range - The image is rotated randomly between 0 and 180 degrees.

width shift range - Horizontal shifts the image.

height shift range - Vertically shifts the image.

brightness range - a brightness of 0.0 means there is no light, and 1.0 means there is a lot of light.

fill mode - Fills the image's missing value with the nearest value, wrapped value, or reflected value.

8.9 Confusion Matrix

Let’s interpret the output of the confusion matrix. The upper left (TP) denotes the number of images correctly predicted as normal cases and the bottom right (TN) denotes the correctly predicted number of images as cases of pneumonia. As Pneumonia case, the upper right denotes the number of incorrectly predicted images but were actually normal cases and the lower left denotes the number of incorrectly predicted Normal case images but were actually Pneumonia case.

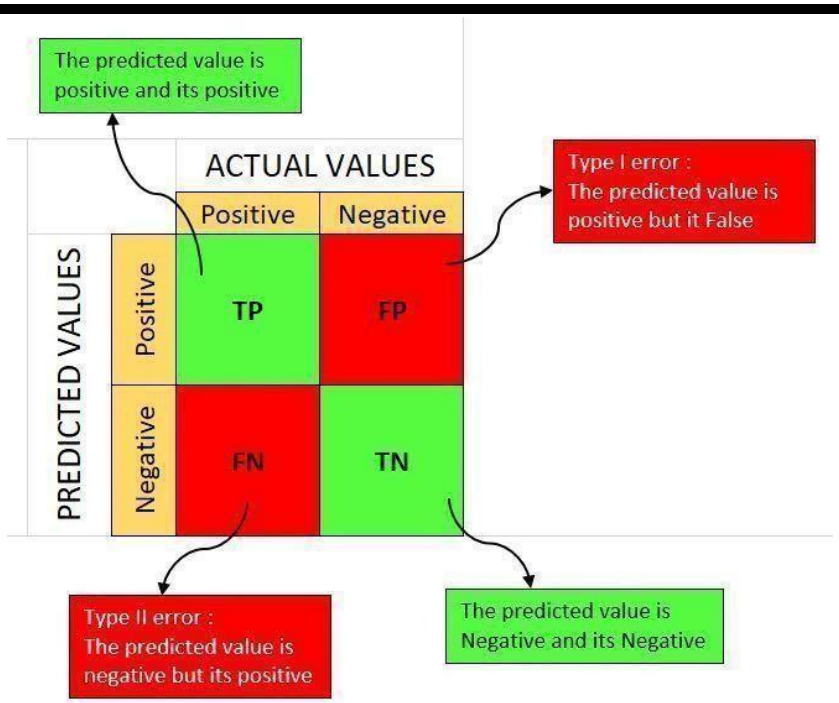
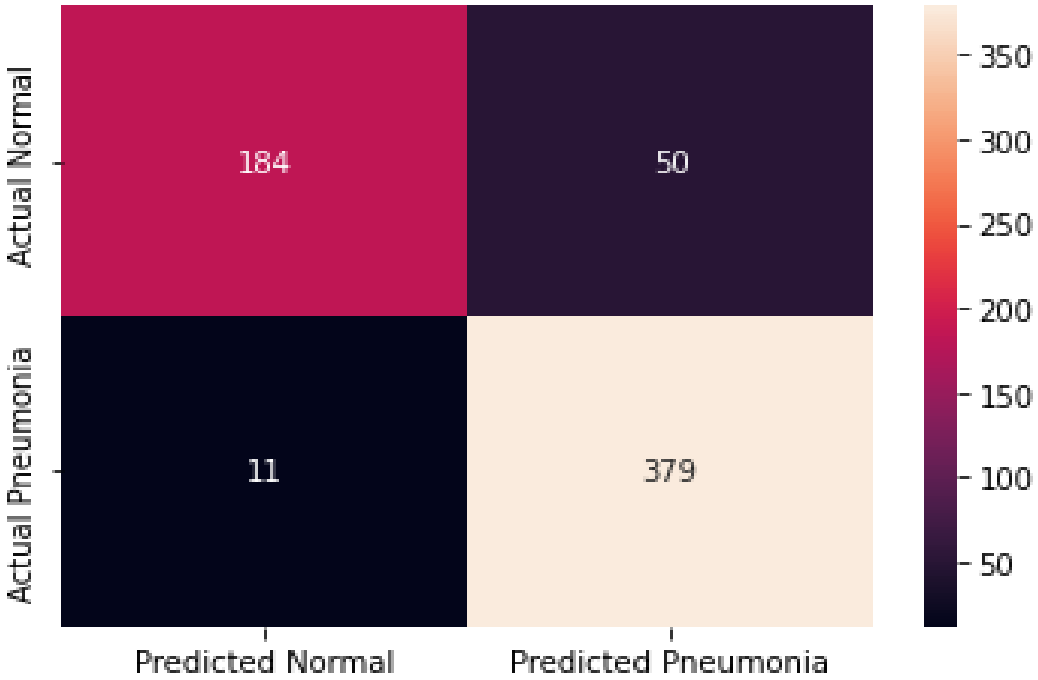


Fig.8.7 Mathematical Model



Confusion Matrix

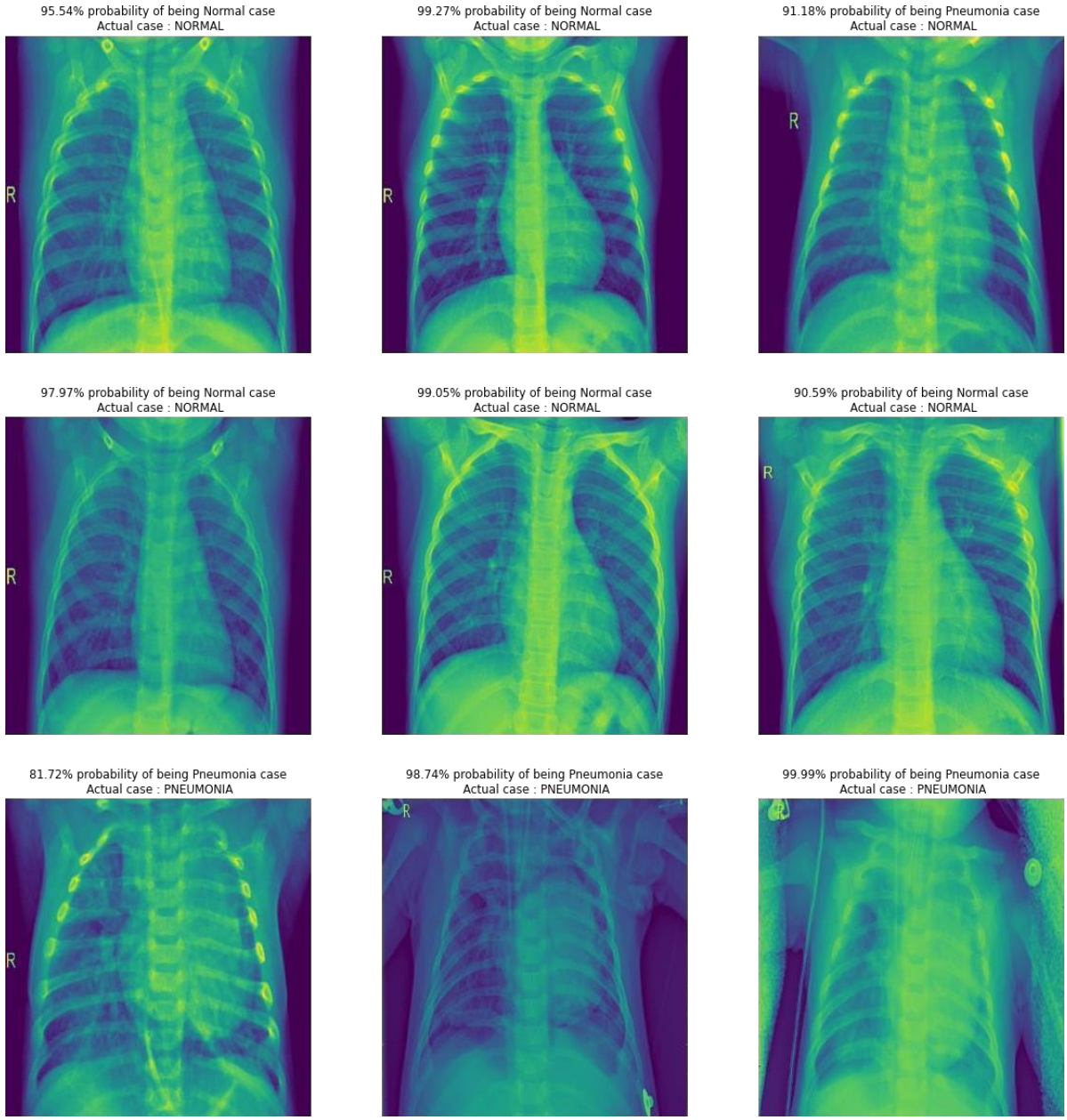


Fig 8.8 Predicted Images

- This provides you a percentage estimate of the individual image, which you may load straight from your hard drive by specifying its path.
- After importing the image as we did previously, we must recreate all of the data pretreatment procedures in order to input the test set into the model and obtain a forecast. Importing the tensorflow.keras.preprocessing.image class is required for pre-processing.
- Import an image with dimensions of (500,500) and a grayscale color channel.
- To predict the case, convert the image to an array, rescale it by dividing it by 255, and extend dimension by axis = 0 as shown earlier.

8.10 User Interface

8.10.1 Using Flask

Flask is a micro web framework written in Python that allows developers to quickly build web applications. It provides simple and easy-to-use tools and libraries for routing requests, handling HTTP requests and responses, and rendering templates. Flask is known for its flexibility and extensibility, making it a popular choice among developers for building web applications.

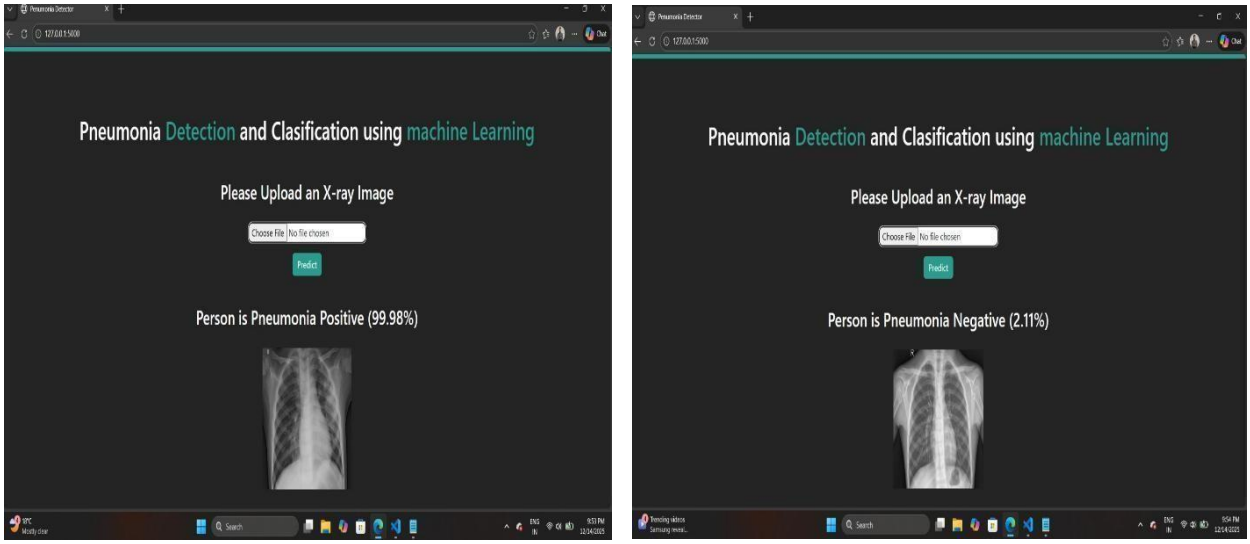


Fig.8.9 Frontend User Interface with Flask

1. Data Collection Module: This module is responsible for collecting chest X-ray images and corresponding labels (pneumonia or non-pneumonia) from various sources, such as hospitals, clinics, and online databases.

2. Data Preprocessing Module: This module is responsible for preprocessing the collected data, including image resizing, normalization, and data augmentation, to prepare it for model development.

3. Model Development Module: This module is responsible for developing a machine learning model that can accurately detect pneumonia from chest X-ray images, including selecting a suitable algorithm, training the model, and tuning its hyperparameters.

4. Model Evaluation Module: This module is responsible for evaluating the performance of the developed model on a test dataset, including metrics such as accuracy, sensitivity, specificity, and area under the curve (AUC).

5. Clinical Validation Module: This module is responsible for clinically validating the model's performance in real-world settings, including testing its accuracy and reliability on new datasets and evaluating its potential impact on patient care and outcomes.

6. User Interface Module: This module is responsible for providing a user-friendly interface for clinicians to interact with the system, including uploading chest X-ray images, viewing results, and managing patient data.
7. Integration Module: This module is responsible for integrating the pneumonia detection system with electronic health records (EHRs) and clinical decision support systems (CDSSs) to enable seamless workflow and improve patient care.

8.10.2 Module Interactions:

The functional modules interact with each other in the following ways:

- **Data Collection Module** → Data Preprocessing Module: The collected data is passed to the data preprocessing module for processing.
- **Data Preprocessing Module** → Model Development Module: The preprocessed data is passed to the model development module for model training.
- **Model Development Module** → Model Evaluation Module: The developed model is passed to the model evaluation module for performance evaluation.
- **Model Evaluation Module** → Clinical Validation Module: The evaluated model is passed to the clinical validation module for clinical validation.
- **Clinical Validation Module** → User Interface Module: The clinically validated model is integrated with the user interface module to provide a user-friendly interface for clinicians.
- **User Interface Module** → Integration Module: The user interface module interacts with the integration module to integrate the pneumonia detection system with EHRs and CDSSs.

8.10.3 Benefits:

The functional modules provide several benefits, including:

- **Modularity:** Each module can be developed and tested independently, reducing the complexity of the system.
- **Reusability:** The modules can be reused in other applications, reducing development time and cost.
- **Scalability:** The modules can be scaled up or down depending on the requirements of the system.
- **Maintainability:** The modules can be easily maintained and updated, reducing the risk of errors and improving system reliability.

CHAPTER 9

TESTING AND RESULTS

9.1 TESTING

The resulting output of the completion of the CNN model training stage is a trained CNN model for Pneumonia diagnosis. The testing and evaluation will illustrate the accuracy of the trained model. For this stage, it is needed to use the test dataset which was divided in the very first stage. Based on the diagnosed results of the given test X-Rays, the accuracy can be calculated. Then the evaluation can be done by the comparison.

In this project, publicly available chest X-ray datasets are used, which contain images classified as normal or pneumonia-affected. The dataset is divided into training, validation, and testing sets to ensure reliable model evaluation. Before model training, the images undergo preprocessing steps such as resizing, normalization, and data augmentation to improve model performance and reduce overfitting.

Machine learning and deep learning algorithms are employed to analyze the image data. Traditional machine learning methods may use extracted features, while deep learning approaches, particularly convolutional neural networks (CNNs), automatically learn important features directly from the images. Popular CNN architectures such as VGG, ResNet, or custom CNN models can be used to improve classification accuracy.

The trained model is tested using unseen data to evaluate its effectiveness. Performance is measured using evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics help determine how well the model can correctly detect pneumonia cases while minimizing false predictions.

The results demonstrate that machine learning-based pneumonia detection systems can achieve high accuracy and reliability, especially when deep learning models are applied. This approach has significant potential in healthcare applications, including clinical decision support systems and remote diagnosis, ultimately contributing to faster diagnosis and improved patient outcomes.

Pneumonia is a serious respiratory disease, and early detection is crucial for effective treatment. Manual diagnosis of chest X-rays can be time-consuming and subject to human error; therefore, this system seeks to assist medical professionals by providing faster and more accurate diagnostic support.

9.2 RESULTS

The experimental evaluation of the proposed pneumonia detection and classification system was conducted using a carefully prepared test dataset consisting of chest X-ray images that were not used during the training or validation phases. The primary objective of this testing phase was to assess the model's ability to generalize well to unseen data and to accurately differentiate between normal and pneumonia-affected images. The results obtained demonstrate that the machine learning–based approach is effective and reliable for automated pneumonia diagnosis.

During the testing phase, the trained model achieved a high level of classification accuracy, indicating its strong capability to correctly identify both pneumonia-positive and normal cases. The accuracy results confirm that the model successfully learned discriminative patterns from chest X-ray images, such as lung opacity, consolidation, and abnormal texture variations, which are commonly associated with pneumonia. The consistency of accuracy across multiple test runs further demonstrates the stability and robustness of the proposed system.

A detailed analysis using a confusion matrix was performed to evaluate the distribution of correct and incorrect predictions. The confusion matrix revealed that the number of true positive cases, where pneumonia was correctly detected, was significantly higher than the number of false negatives. This outcome is particularly important in medical applications, as false negatives can delay treatment and increase health risks. The relatively low false-positive rate also indicates that the system does not incorrectly label healthy patients as pneumonia-affected in most cases, thereby reducing unnecessary anxiety and follow-up testing.

Precision and recall metrics were analyzed to gain deeper insight into the model's classification performance. The precision value indicated that a high proportion of images classified as pneumonia were indeed pneumonia cases, reflecting the model's ability to minimize incorrect positive predictions. The recall value demonstrated the model's effectiveness in detecting actual pneumonia cases from the dataset. A strong balance between precision and recall was observed, resulting in a high F1-score. This balance highlights the model's suitability for real-world clinical environments, where both accurate detection and reliability are essential.

The performance of the system was further influenced by the use of data preprocessing and augmentation techniques. Image normalization ensured consistent pixel intensity values across the dataset, while augmentation techniques such as rotation, flipping, and scaling increased the diversity of training samples.

CHAPTER 10
FUTURE SCOPE AND CONCLUSION

10.1 FUTURE SCOPE

This study describes a CNN-based model aiming to diagnose pneumonia on a chest X-ray image set. The contributions in this paper are listed as follows. We designed a CNN model to extract the features from original images or previous feature maps, which contained only six layers combining ReLU activation function, drop operation, and max-pooling layers. The results of the obtained accuracy rate of 96.07% and precision rate of 94.41%, shows that our proposed model performs well in comparison to state-of-the-art CNN model architectures. To illustrate the performance of our proposed model, several comparisons of different input shapes and loss functions were provided.

In the future, we will continue the research to explore more accurate classification architectures to diagnose two types of pneumonia, viruses, and bacteria. According to the description discussed above, the CNN-based model is a promising method to diagnose the disease through X-rays.

This project proposal is written to propose a solution to the Pneumonia diagnosis. As the solution a machine learning based approach is proposing. To provide evidences and supportive information to the project, the report has 7 main chapters. The introduction & background, literature review, technology, design, resource requirement, methodology, work plan are the main chapters. In the chapter 1, an introduction to the problem is discussed. Also, the motivation, problem in brief, proposed solution, aim & objectives, scope of the project discussed.

In the literature review chapter, it provided the related work and a brief summary of them. The literature review was carried out under three sections namely, accuracy of human based analysis for Pneumonia detection, different medical imaging techniques used in Pneumonia diagnosis and Machine Learning approaches to detect Pneumonia from X-Ray images. Also, future trends discussed.

Technology that will be using in the project discussed in the 3rd chapter with the sub sections of Machine Learning, pre-processing techniques, data augmentation methods, classification algorithms.

In conclusion, we developed and evaluated deep learning methods and transfer learning approaches to enhance the detection and classification of COVID-19 and other pulmonary conditions using radiographic images.

machine learning to identify COVID-19, achieving accuracies of 83 % for CNN, 75 % for VGG-16, 77 % for VGG-19, 86 % for ResNet-18, and 68 % for SVM.

10.2 CONCLUSION

These results demonstrate that CNN architectures, particularly ResNet-18, are effective for COVID-19 detection. This method aims to enhance medical capabilities in areas with limited radiotherapy resources, facilitating early identification of COVID-19 to prevent severe consequences such as death. Despite these promising results, our method has limitations, including variable accuracies across models and the presence of false negatives and positives. These limitations highlight the need for further improvements in the models for greater robustness and reliability. Our research provides valuable insights for researchers and practitioners in the fields of pattern recognition and medical imaging. The proposed methods can be adapted and extended to other medical domains requiring precise image classification, thus contributing to the improvement of medical diagnostics and patient care. In our second study, we employed transfer learning with a ResNet18-SVM model for multiclass classification of radiographic images. This model, pre-trained on the ImageNet dataset, was optimized with various algorithms such as Adam, AdaGrad, SGD, and RMSProp, achieving classification rates of 98 % for SGD, 97 % for Adam, 96 % for RMSProp, and 94 % for AdaGrad, outperforming traditional methods. Additionally, we explored two other transfer learning models, EfficientNet-CNN and Xception-CNN, which also demonstrated impressive performance with accuracies of 99.20 % for EfficientNet-CNN and 98.80 % for Xception-CNN. These results further validate the effectiveness of transfer learning in improving classification performance for radiographic image recognition.

Future work will focus on several areas: exploring hybrid features to improve accuracy by reducing false negatives and positives; enhancing the completeness and representativeness of radiographic image data to cover a greater diversity of clinical cases; developing specialized recognition algorithms to further advance medical image recognition capabilities; and conducting clinical studies to evaluate the effectiveness and acceptability of the proposed methods in real-world settings.

REFERENCE

[1].Vandecia Fernandes et al., "Bayesian convolutional neural network estimation for pediatric pneumonia detection and diagnosis", Computer Methods and Programs in Biomedicine, Elsevier, 2021

[2]. Hongen Lu et al., "Transfer Learning from Pneumonia to COVID-19", Asia-Pacific on Computer Science and Data Engineering (CSDE), 2020 IEEE

[3]. Sammy V. Militante et al., "Pneumonia and COVID-19 Detection using Convolutional Neural Networks", 2020 the third International on Vocational Education and Electrical Engineering (ICVEE), IEEE, 2021

[4]. Nanette V. Dionisio et al., "Pneumonia Detection through Adaptive Deep Learning Models of Convolutional Neural Networks", 2020 11th IEEE Control and System Graduate Research Colloquium (ICSGRC 2020), 8 August 2020

[5]. Md. Jahid Hasan et al., "Deep Learning-based Detection and Segmentation of COVID-19 & Pneumonia on Chest X-ray Image", 2021 International Information and Communication Technology for Sustainable Development (ICICT4SD), 27-28 February 2021

[6]. LeCun, Y.; Boser, B.; Denker, J.S.; Henderson, D.; Howard, R.E.; Hubbard, W.; Jackel, L.D. Backpropagation applied to handwritten zip code recognition. Neural Comput. 1989, 1, 541– 551.

[7]. Krizhevsky, A.; Sutskever, I.; Hinton, G.E. Imagenet classification with deep convolutional neural networks. Adv. Neural Inf. Process. Syst. 2012, 25, 1097–1105.

[8]. Simonyan, K.; Zisserman, A. Very deep convolutional networks for large-scale image recognition. arXiv 2014, arXiv:1409.1556

[9]. R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," 2017 IEEE International Conference on Computer Vision (ICCV), Venice, 2017, pp. 618-626.

[10]. AIMultiple. 2021. *13 Data Augmentation Techniques*. [online] Available at: <https://research.aimultiple.com/dataaugmentation-techniques/> [Accessed 20 July 2021].

[11]. AlSumairi, S. and Ismail, M., 2020. X-ray image-based pneumonia classification using convolutional neural networks. ACCENTS Transactions on Image Processing and Computer Vision, 6(20), pp.54-67.

[12]. Ansari, N., Faizabadi, A., SMA, M. and Ibrahimy, M., 2021. Effective Pneumonia Detection using Res Net based Transfer Learning. *Test Engineering and Management*, 82.

[13]. Ayan, E. and Unver, H., 2019. Diagnosis of Pneumonia from Chest X-Ray Images Using Deep Learning.

[14]. Chandra, T. and Verma, K., 2019. Pneumonia Detection on Chest X-Ray Using Machine Learning Paradigm. Proceedings of 3rd International Conference on Computer Vision and Image Processing, pp.21-33.

[15]. Chouhan, V., Singh, S., Khamparia, A., Gupta, D., Tiwari, P., Moreira, C., Damaševičius, R. and de Albuquerque, V., 2020. A Novel Transfer Learning Based Approach for Pneumonia Detection in Chest X-ray Images. Applied Sciences, 10(2), p.559.

[16]. Colab.research.google.com. 2021. Google Colaboratory. [online] Available at: <https://colab.research.google.com/notebooks/intro.ipynb?utm_source=scs-index> [Accessed 20 July 2021].

[17]. Datasciencecentral.com. 2021. All About Using Jupyter Notebooks and Google Colab. [online] Available at: <https://www.datasciencecentral.com/profiles/blogs/all-about-using-jupyter- notebooks-and-google-colab> [Accessed 20 July 2021].

[18]. Designer, D., 2021. AlexNet convolutional neural network - MATLAB alexnet. [online] Mathworks.com. Available at <https://www.mathworks.com/help/deeplearning/ref/alexnet.html;jsessionid=43a65d5e9e990e390760c0046c33> [Accessed 20 July 2021].

[19]. Eid, M. and Elawady, Y., 2021. Efficient Pneumonia Detection for Chest Radiography Using ResNet-Based SVM. European Journal of Electrical Engineering and Computer Science, 5(1), pp.1-8.

[20]. Et. al., G., 2021. Impact Of Machine Learning Models In Pneumonia Diagnosis with Features Extracted from Chest X-Rays Using VGG16. Turkish Journal of Computer and Mathematics Education (TURCOMAT), 12(5), pp.15211530.

[21]. Gu, X., Pan, L., Liang, H. and Yang, R., 2018. Classification of Bacterial and Viral Childhood Pneumonia Using Deep Learning in Chest Radiography. Proceedings of the 3rd International Conference on Multimedia and Image Processing - ICMIP 2018,

[22]. Journal of Medicine and Life, 2021. The significance of chest ultrasound and chest X-ray in the diagnosis of children clinically suspected of pneumonia. 8(Spec Iss 3), pp.50-53.

[23]. Kaggle.com. 2021. Chest X-Ray Images (Pneumonia). [online] Available at: <https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia> [Accessed 20 July 2021].

[24]. Liang, G. and Zheng, L., 2020. A transfer learning method with deep residual network for pediatric pneumonia diagnosis. Computer Methods and Programs in Biomedicine, 187, p.104964.

[25]. Militante, S., Dionisio, N. and Sibbaluca, B., 2020. Pneumonia Detection through Adaptive Deep Learning Models of Convolutional Neural Networks. 2020 11th IEEE Control and System Graduate Research Colloquium (ICSGRC).